

Datum Atlas Guide

A materials workbench: choose materials, check their life and how to make them, and see their structure, step by step.

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github.com/Edgar-Mendonca/Datum-Atlas

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Welcome

Datum Atlas is a materials workbench: modules that share one library of referenced materials. Today they are **Library**, **Select**, **Eco**, **Make**, **Microstructure**, **Crystal**, **Testing**, **EBSD** and **DIC**; image analysis, EBSD, DIC, corrosion and heat treatment are on the way.

The Select module helps you choose a material and show why it is the right one. Every material is drawn on property charts as a bubble spanning its range of values; design goals and performance indices turn what the part must do into a line on the chart; limits, lines, boxes and shapes narrow the field; and the materials that pass are ranked, with the reason every other one fails. Then you compare the best, add up their energy and CO_2 through the product's life, find the processes that can make the part and what it costs, and print a report that documents the choice.

This guide takes you through Datum Atlas step by step. Each chapter stands on its own.

How to read this guide:

- Names in **bold** are buttons, tabs and boxes exactly as they appear in Datum Atlas.
- An arrow › is a path: **Select › Design Goal** means the **Select** tab, then the **Design Goal** button.
- Keys are shown like this: `Ctrl+K`.
- The orange numbers on the pictures match the numbered lists under them.

> **Tip.** Almost every tool has a small **?** beside it. Click it for a card that says what the tool does and how to use it. **Help › What Is This Tool?** lists them all.

About the data

Every value in Datum Atlas is compiled from open and public sources: government handbooks, open datasets, university notes, producers' own datasheets and typical textbook ranges. Each value names its source, and a ✓ marks values checked against a published datasheet; the others are drafts. A value that is not known is a gap, never a zero. The values describe generic materials: before you decide, check the supplier's data for the grade and condition you will use.

Install, update and start

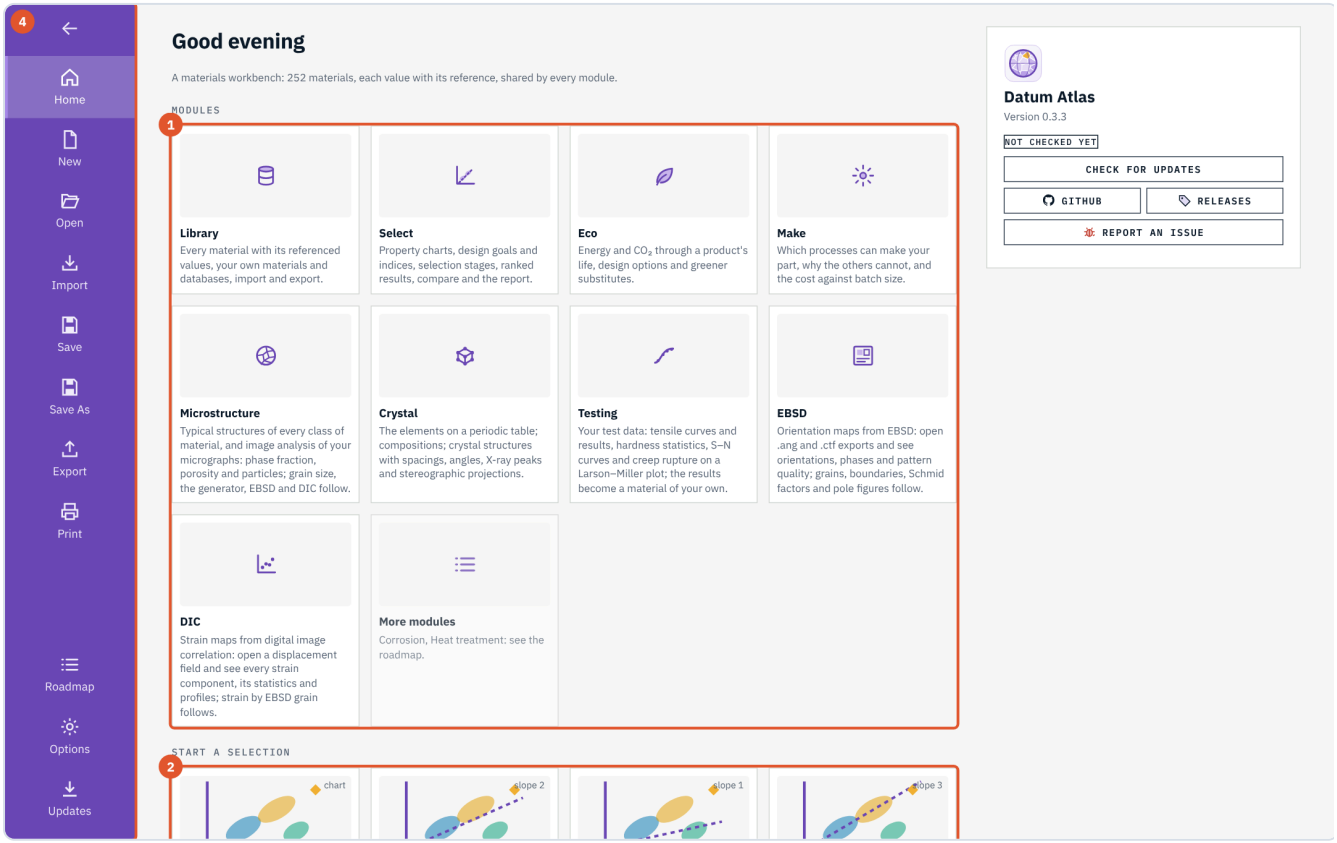
Install

1. Download `DatumAtlas-x.y.z-setup.exe` from the Releases page: github.com/Edgar-Mendonca/Datum-Atlas/releases.
2. Run it. It installs for your own Windows account, so you do not need administrator rights.
3. The first time, Windows may show **Windows protected your PC**, because the installer is not code-signed yet. Click **More info**, then **Run anyway**.

Updates

Datum Atlas checks for a new version when it starts. When one is ready, **File › Updates** shows it; click **Update now**. The download is checked against its checksum before it installs, and your projects and materials are not touched.

The Home page



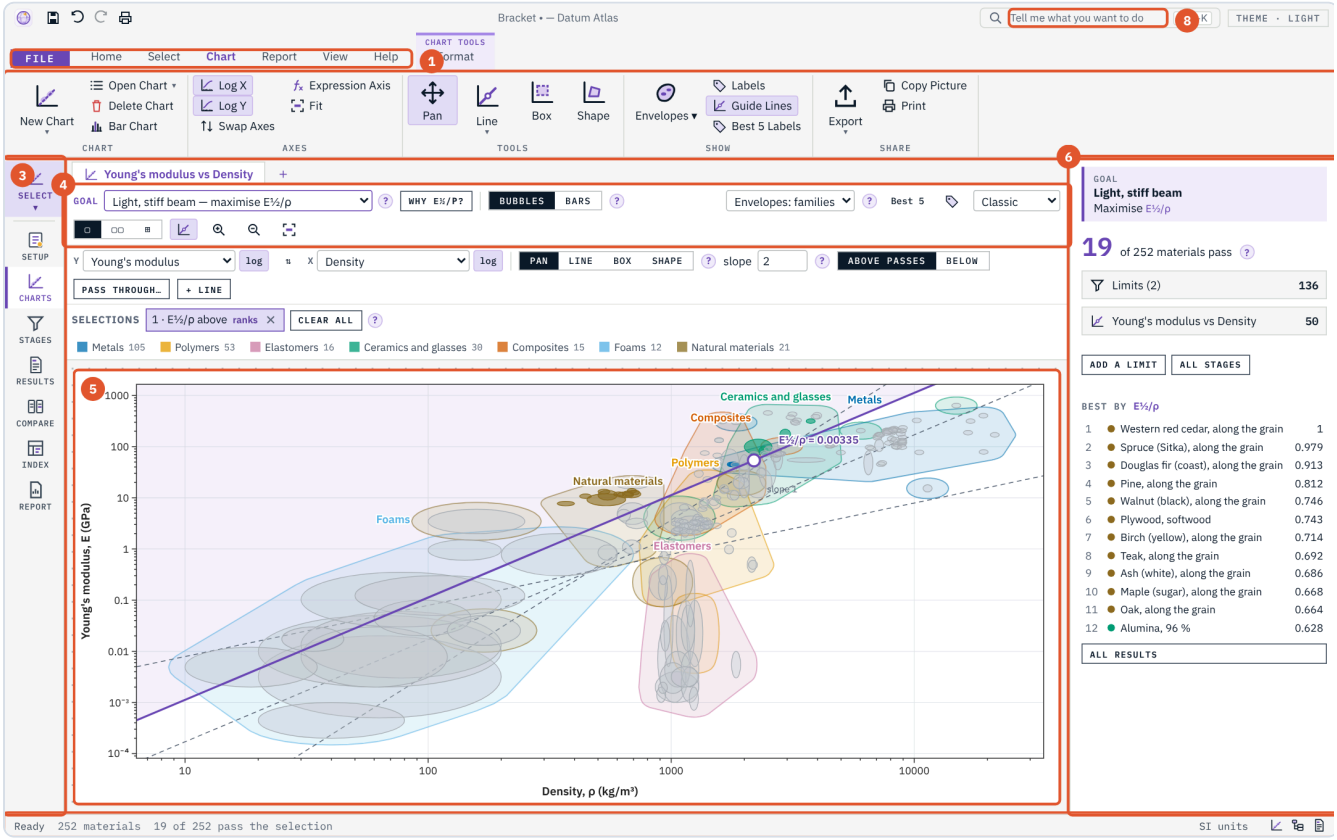
The Home page (File › Home).

1. **Modules.** A tile per module opens it; **More modules** shows the ones on the way in the roadmap.
2. **Start a selection.** Each design problem starts a project set up for a classic design: a light, stiff beam; a light, strong tie; a stiff panel; springs; a hot, stiff and light part; leak before break. **Blank selection** starts with one chart and nothing selected.

3. **Add a material**, the **Guide** and **Open** are here too.
4. **The File page** also has **New**, **Open**, **Import**, **Save**, **Save As**, **Export**, **Print**, **Roadmap**, **Options** and **Updates**. The arrow at the top goes back to your project.

Projects are saved as `.datlas` files. Your own materials live in a database of their own (`.datdb`), so every project sees them.

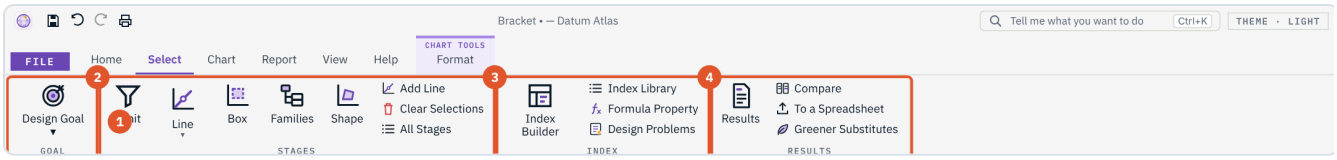
The window at a glance



The Datum Atlas window with a light, stiff beam on Young's modulus against density.

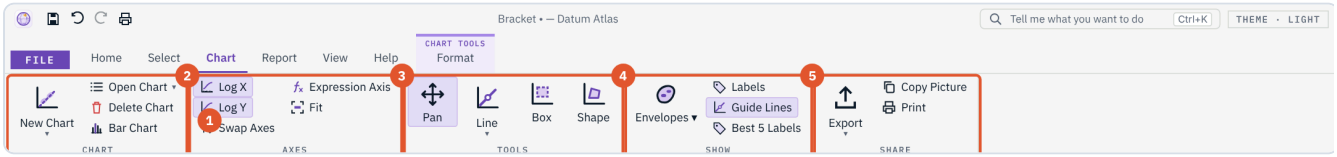
1. **File and the ribbon tabs: Home**, **Select**, **Chart**, **Data**, **Report**, **View** and **Help**. **Chart Tools › Format** appears while a chart is open.
2. **The ribbon.** The buttons of the chosen tab, in named groups.
3. **The sidebar.** At the top, the open module (click it to switch); below, its pages: here the Select module's Setup, Charts, Stages, Results, Compare, Index and Report.
4. **The chart bar.** The design goal, bubbles or bars, envelopes, the five best labelled, the chart style, the axes and the chart tools.
5. **The property chart.** Every material as a bubble; hover for its values, click to pick it, double-click for its datasheet.
6. **The stage panel.** How many materials pass, each stage with its count, and the best materials by the ranking.
7. **The status bar.** The number of materials, how many pass, the material picked, and **SI units** or **Imperial units** (click to switch).
8. **Tell Me** (`Ctrl+K`). Type what you want to do, a design goal or a material, and press `Enter`.

The ribbon at a glance:



The Select tab.

1. **Goal:** pick a design goal for the open chart.
2. **Stages:** add a limit, a selection line, a box, a shape, leave families out; add a second line; clear the selections; see all stages.
3. **Index:** the index builder and library, your own formula properties, the design problems.
4. **Results:** the ranked results, Compare, results to a spreadsheet, greener substitutes.



The Chart tab.

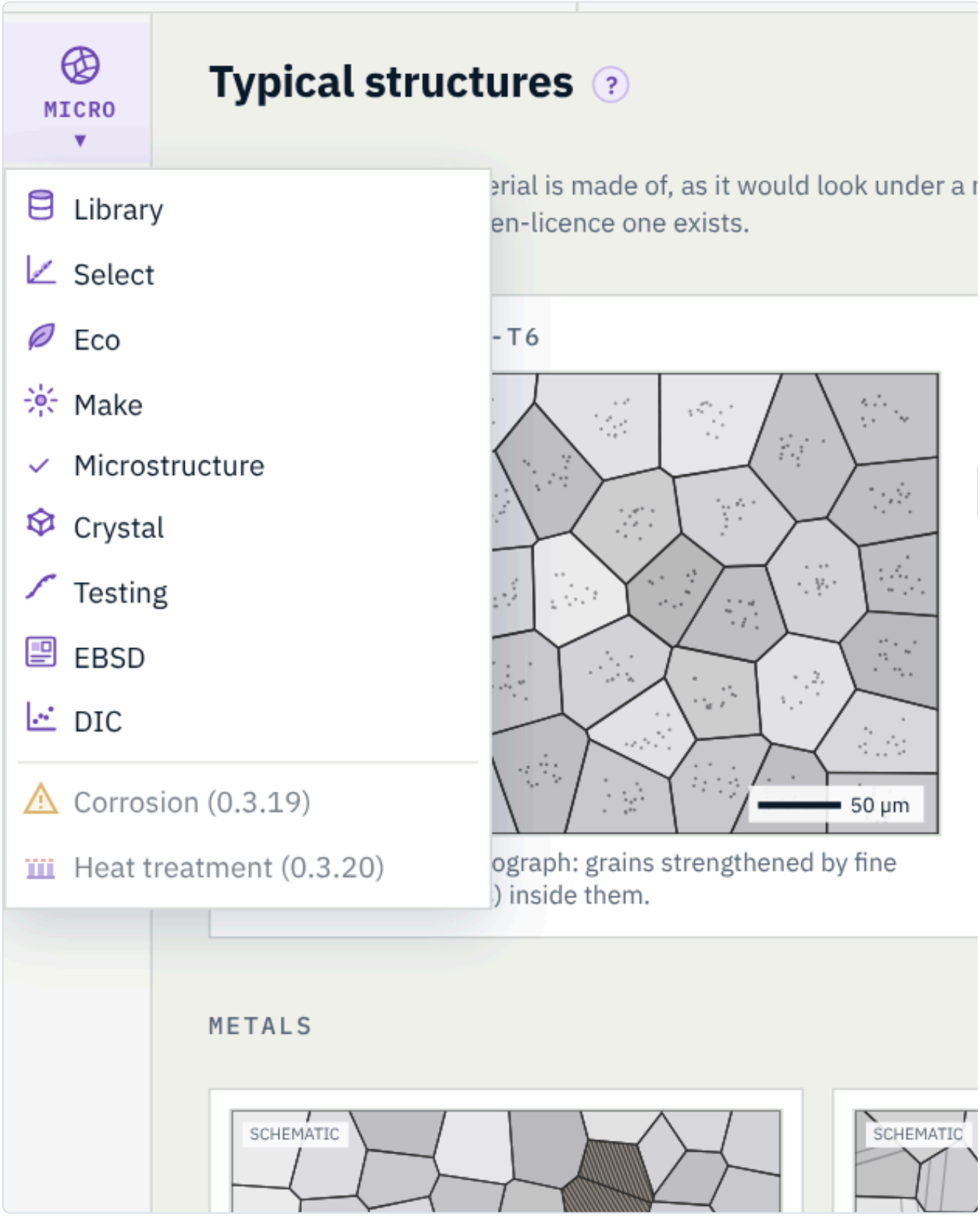
1. **Chart:** a new chart, open or delete one, a bar chart of one property.
2. **Axes:** log or linear scales, swap the axes, an expression axis (a formula), fit to the data.
3. **Tools:** what a drag on the chart does: pan, draw a line, a box or a shape.
4. **Show:** envelopes, labels, guide lines, the best five labelled.
5. **Share:** export the chart as a picture (SVG or PNG), copy it, print it.



The Report tab.

1. **Compare** and the radar chart.
2. **Report:** the selection report, print, the results as a spreadsheet or CSV.
3. **Eco:** the eco audit, its charts and greener substitutes.
4. **Process:** process selection, the process records and the cost model.

Modules



The module switcher at the top of the sidebar.

Click the module at the top of the sidebar to switch. Each module shows its own pages in the sidebar and its own tabs on the ribbon; **File**, **View** and **Help** are always there. One project file holds the work of every module.

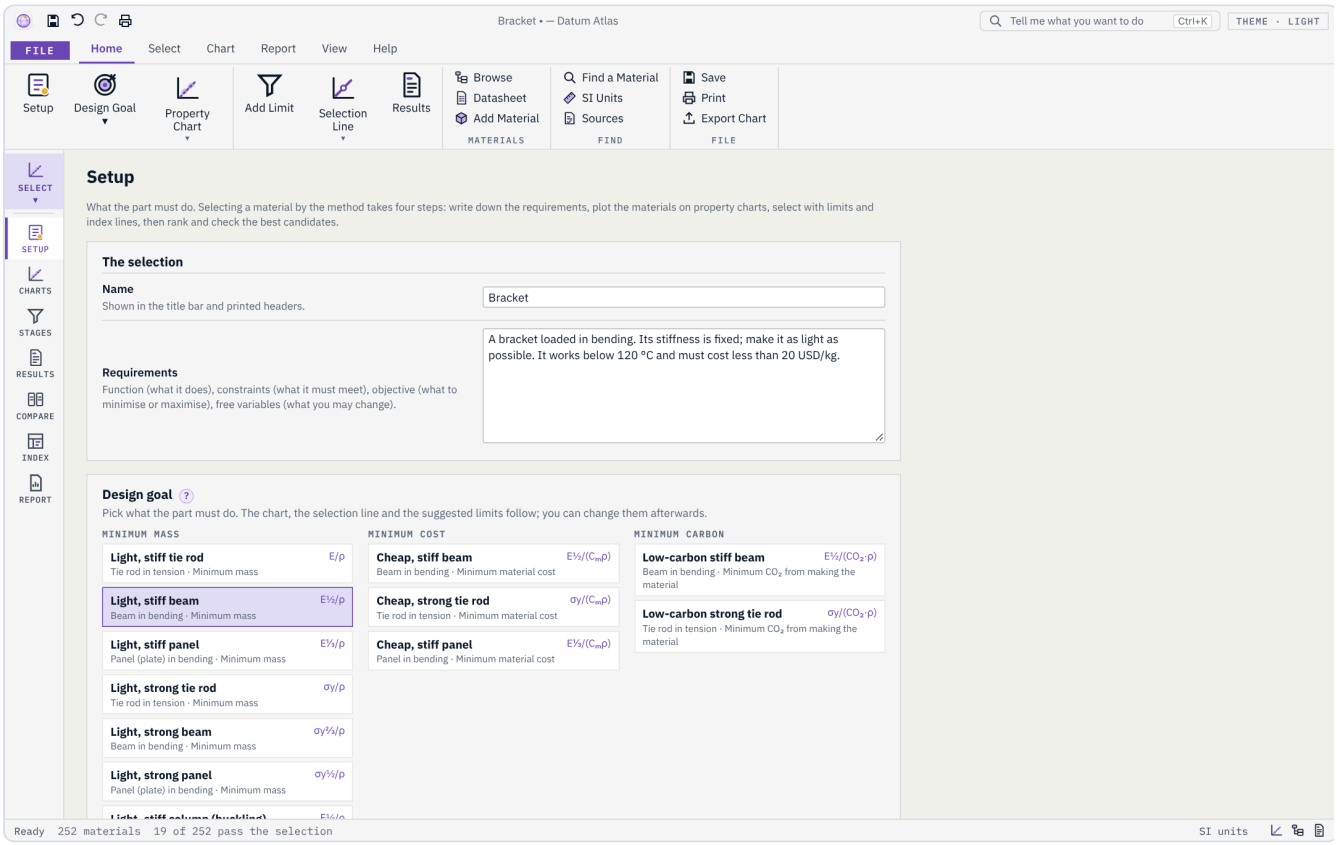
Module	Pages	What it is for
Library	Browse, Datasheet, Data, Sources	Every material with its referenced values; your own materials and databases; import and export
Select	Setup, Charts, Stages, Results, Compare, Index, Report	Choosing a material: charts, goals and indices, stages, ranking, compare, the report
Eco	Eco audit	Energy and CO_2 through a product's life; greener substitutes
Make	Process, Processes	Which processes can make the part, and the cost
Microstructure	Structures, Images	The typical structure of every class of material; image analysis of your micrographs
Testing	Tensile, Hardness, Fatigue, Creep	Your test data: tensile results with statistics, hardness statistics, S–N curves, creep rupture; the results become a material of your own
DIC	Strain maps	Strain maps from displacement fields, statistics and profiles
EBSD	Maps, Grains, Pole figures	Orientation maps from EBSD exports; grains, boundaries, twins, KAM, Schmid factors
Crystal	Elements, Lattice	The elements on a periodic table; compositions; crystal structures, spacings, angles, X-ray peaks and stereographic projections

The dimmed modules in the list are on the way; click one to see what it will do and when.

The method in five steps

Datum Atlas follows the method engineers learn for choosing materials:

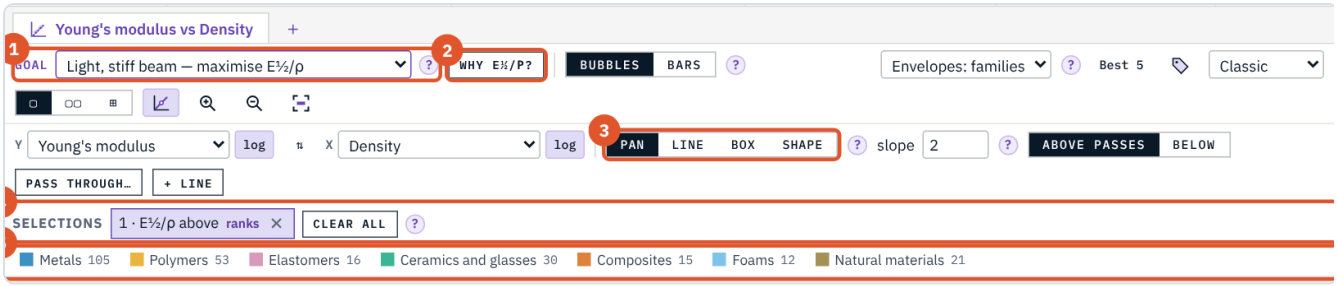
1. **Write down what the part must do.** Its *function* (carry a load in bending, hold pressure, conduct heat), its *constraints* (stiffness, strength, 150 °C, cost), its *objective* (minimise mass, cost or CO_2) and its *free variables* (what you may change: thickness, section, the material). **Setup** has a box for this.
2. **Look at the materials on a property chart.** Two properties on log axes. Each family has an envelope.
3. **Select.** Limits remove what cannot work; a selection line keeps the materials that do the job best.
4. **Rank and check.** Results lists what passes, ranked by the index, and why each other material fails. Open the datasheets of the best few and check their references.
5. **Check its life and how to make it.** The eco audit and process selection.



Setup: the requirements, the design goal and the design problems.

Property charts

A property chart puts every material on two properties: here Young's modulus against density. Each material is an ellipse spanning its range of both values, so a material whose stiffness varies with grade is a tall bubble. Each family (metals, polymers, elastomers, ceramics and glasses, composites, foams, natural materials) has a coloured envelope around its members. Log axes fit materials that differ a thousand times on one chart.



The chart bar and tools.

1. **Design goal.** Sets up the chart for a goal (next chapter).
2. **Why E½/ρ?** Explains the goal's index.
3. **Pan, Line, Box, Shape.** What a drag on the chart does.
4. **Selections.** The lines, boxes and shapes on this chart, as chips: click one to edit it, × to remove it.
5. **The legend.** Click a family to hide or show it on this chart.

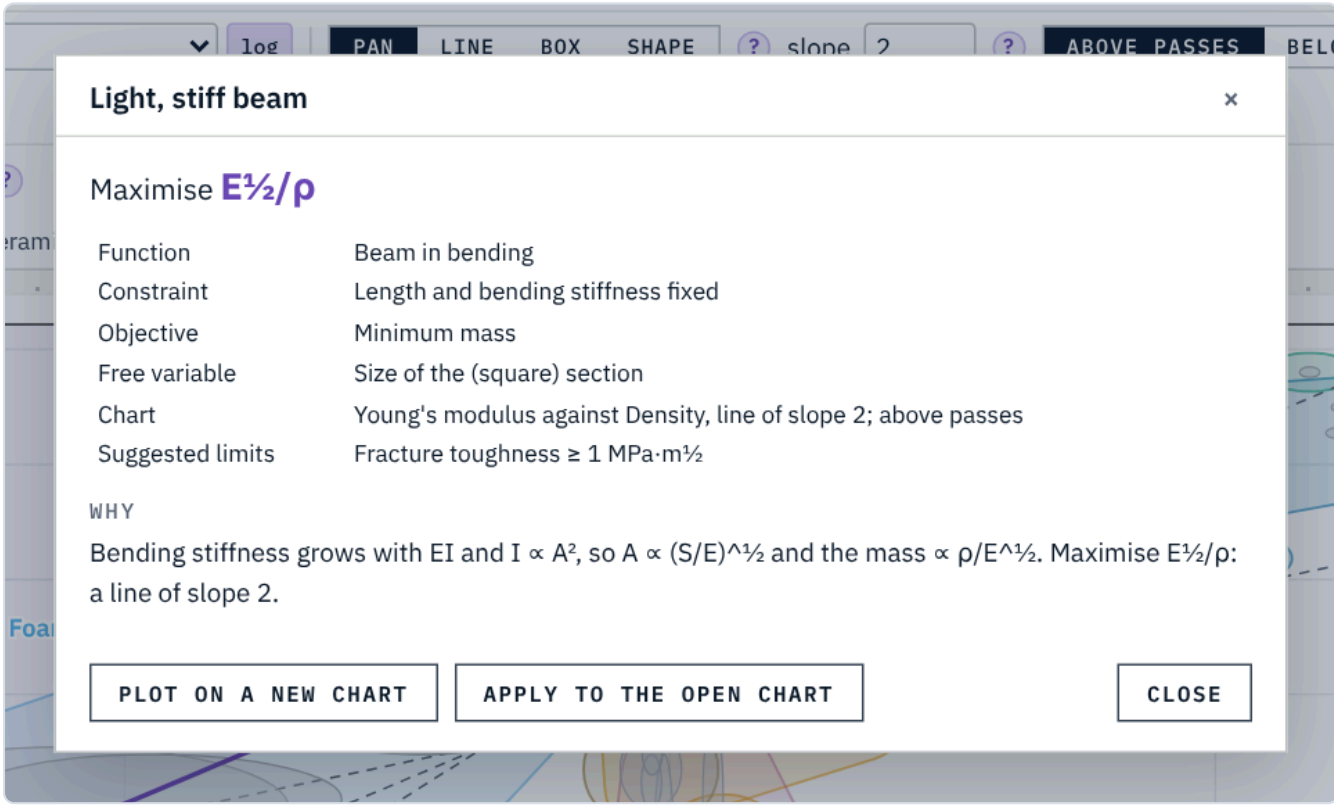
Working on the chart:

- **Hover** a bubble for its values; **click** to pick it (its datasheet shows beside the chart); **double-click** for the full datasheet.
- **Wheel** to zoom around the mouse; drag to pan with **Pan**; **Fit** shows everything.
- **Y** and **X** choose the properties; **log** switches each scale. **Expression Axis** plots a formula, such as `[E]^0.5/[rho]`.
- **Bars** draws one property for every material as a bar from its low to its high value, grouped by family.
- **Guide lines** of slope 1, 2 and 3 show the directions of the classic indices.
- **+** beside the chart tabs adds a chart. The ▢, ▢▢ and ⊞ buttons in the chart bar show one, two or four charts at once, and the material under the mouse is lit on all of them.
- **Chart style:** classic colours, publication (finer lines) or monochrome (greys, for black-and-white print).

> **Tip. Labels** names every material; **Best 5** names only the five best by the ranking. Labels move apart so they do not overlap; drag one to place it, and a leader line follows.

Design goals and performance indices

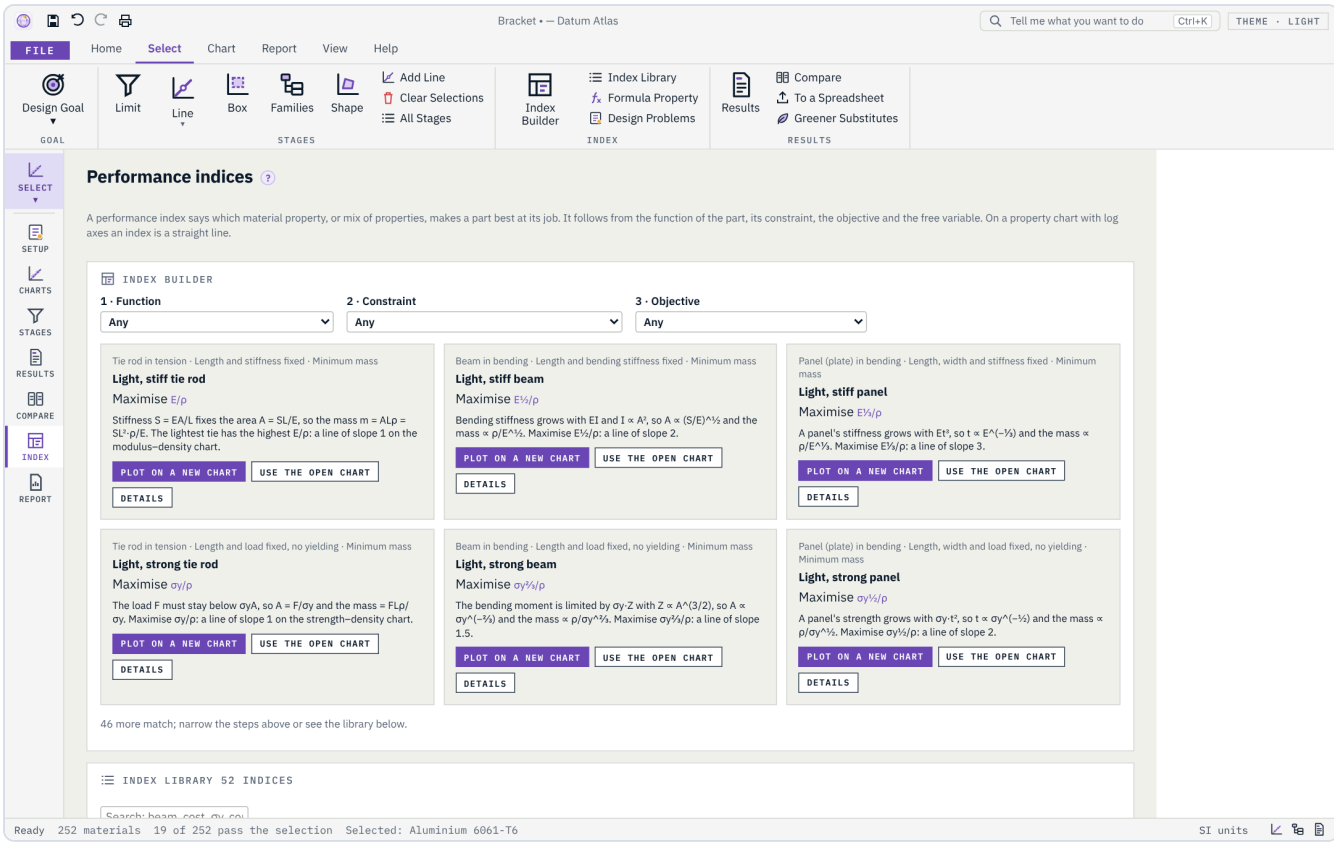
A **design goal** turns what the part must do into a line on a chart. Pick **Light, stiff beam** and Datum Atlas puts Young's modulus against density, draws a selection line of slope 2 through the best materials (the top fifth pass), adds the limits the goal needs (marked **from the goal**) and ranks the results by its index, E½/ρ.



Why E½/ρ? The function, constraint, objective and free variable behind the index.

The **index** follows from the design. A beam of fixed length and bending stiffness, made as light as possible, with its section free: its stiffness grows with E·I and I with the square of the area, so the mass falls as ρ/E½. The lightest beam has the highest E½/ρ: on log axes, a line of slope 2. A tie gives E/ρ (slope 1); a panel gives E⅓/ρ (slope 3).

There are 25 design goals, grouped: minimum mass, minimum cost, minimum carbon, elastic design, damage tolerance, thermal, and two constraints (stiff **and** strong, which set up two charts). **Explore** removes the goal's line and limits; your own limits are never touched.



The Index page: the index builder and the library.

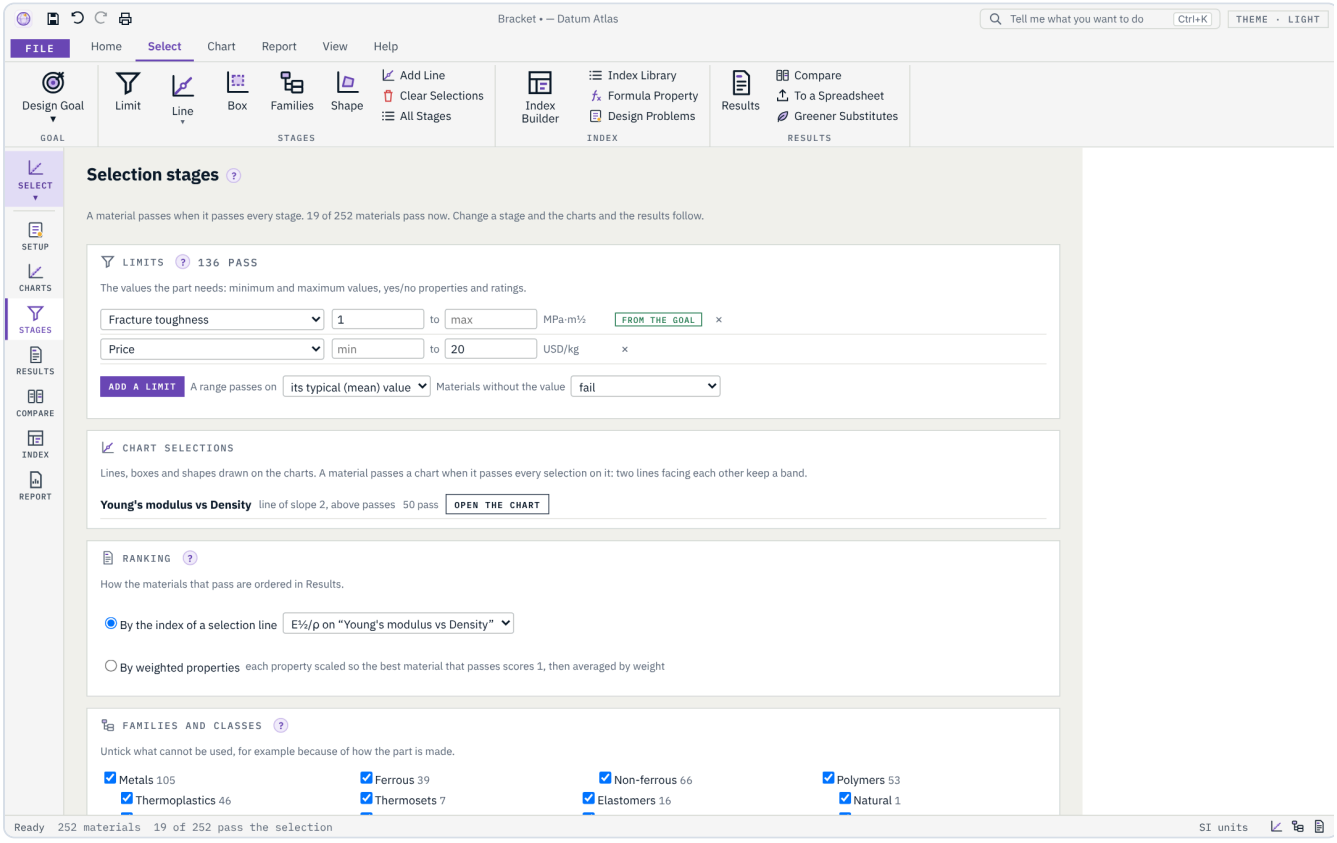
The **Index** page has 52 indices. The **Index builder** narrows them in three steps: the **function** (tie, beam, panel, column, shaft, spring, pressure vessel…), the **constraint** and the **objective**. Each card gives the index and how it follows; **Plot on a new chart** keeps the open chart, **Use the open chart** applies it there.

Your own formula properties

Data › Formula Property defines a property of your own from others, for example `[sy]^2/[E]` for a spring or `[E]*[alpha]` for thermal stress. It is worked out for every material, with its range (low with low, high with high), and works like any other property on charts, in limits and in the ranking. A material without one of the values gets a gap, never a zero.

Selecting

A material passes when it passes **every stage**.



The Stages page: limits, chart selections, ranking, and families.

Limits

Add a limit, choose a property and type a minimum, a maximum or both, in the units shown. A yes/no property (recyclable, combustible…) is a tick; a rating (fresh water, weak acids, UV light, fire resistance, castability, machinability, weldability) takes the lowest rating that will do.

- **A range passes on** its typical (mean) value, or only if the **whole range** passes.
- **Materials without the value** fail, or pass marked **no data**.

Selections on a chart

- **A line** keeps the materials above it (or below). Type its **slope**, or let a goal set it; drag its round handle, nudge it with the arrow keys, or **Pass through…** a value of the index. `Delete` removes the active selection.
- **+ Line** adds a second line. Two lines of the same slope facing each other keep a **band**.
- **Box** keeps a window of values on both properties; **Shape** draws any outline: click the corners, then click the first corner, double-click or press `Enter` to close it.

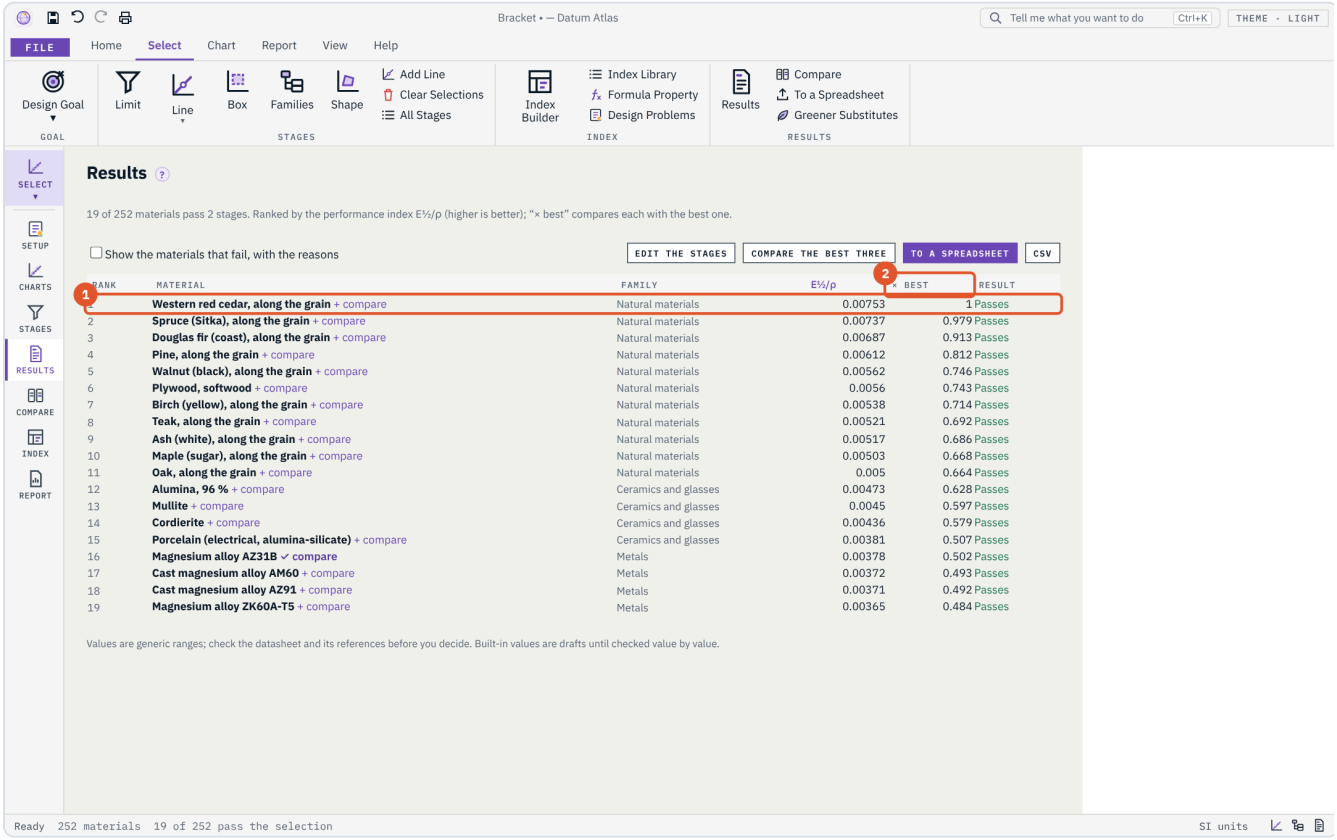
Families

Untick the families and classes that cannot be used: no polymers in a hot part, no wood where it gets wet.

Ranking

The materials that pass are ranked **by the index of a selection line**, or **by weighted properties**: choose properties, whether higher or lower is better, and a weight for each; each property is scaled so the best material that passes scores 1.

Results

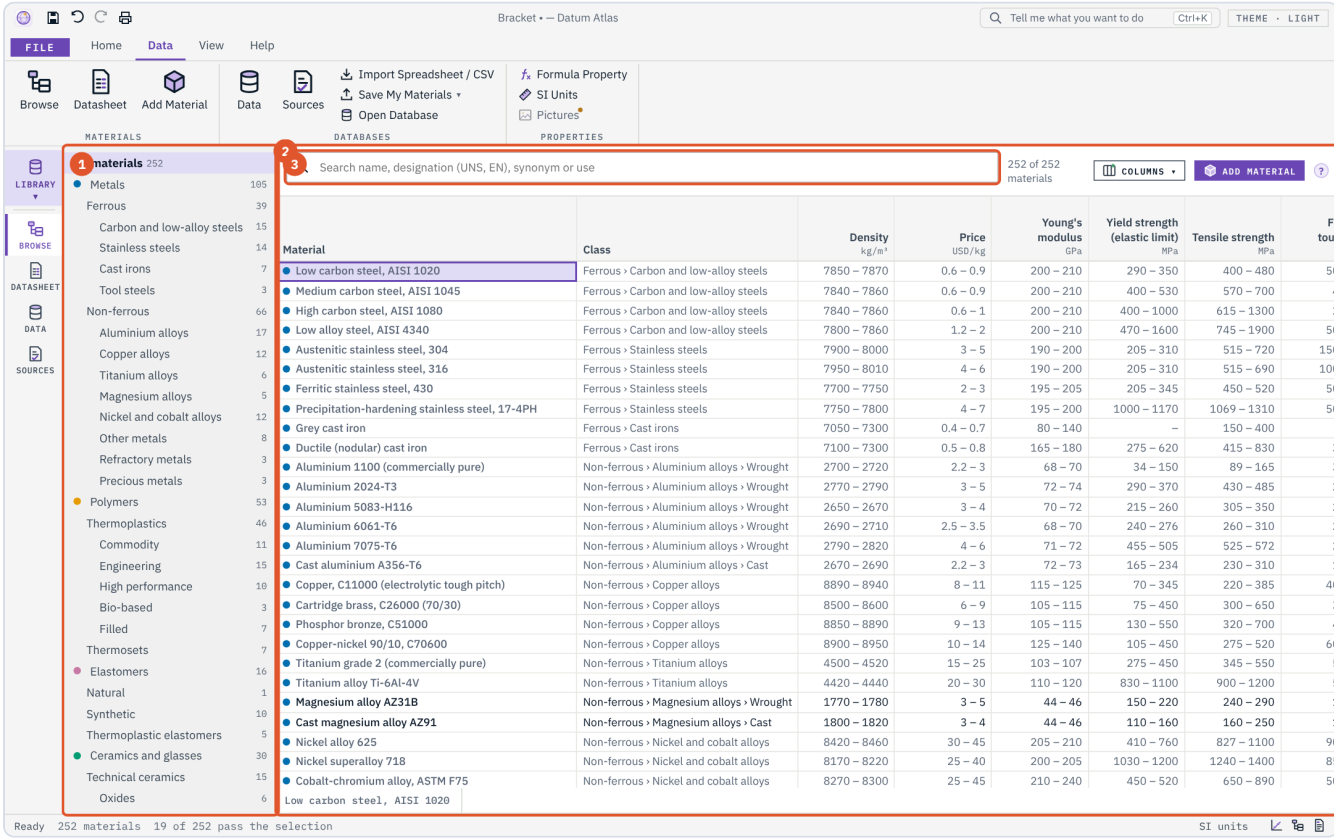


Results: every material ranked, with the reason each other one fails.

1. **The materials that pass**, ranked, with their index value.
2. **× best**: each material's index divided by the best material's: 1 for the best, 0.5 for half as good.

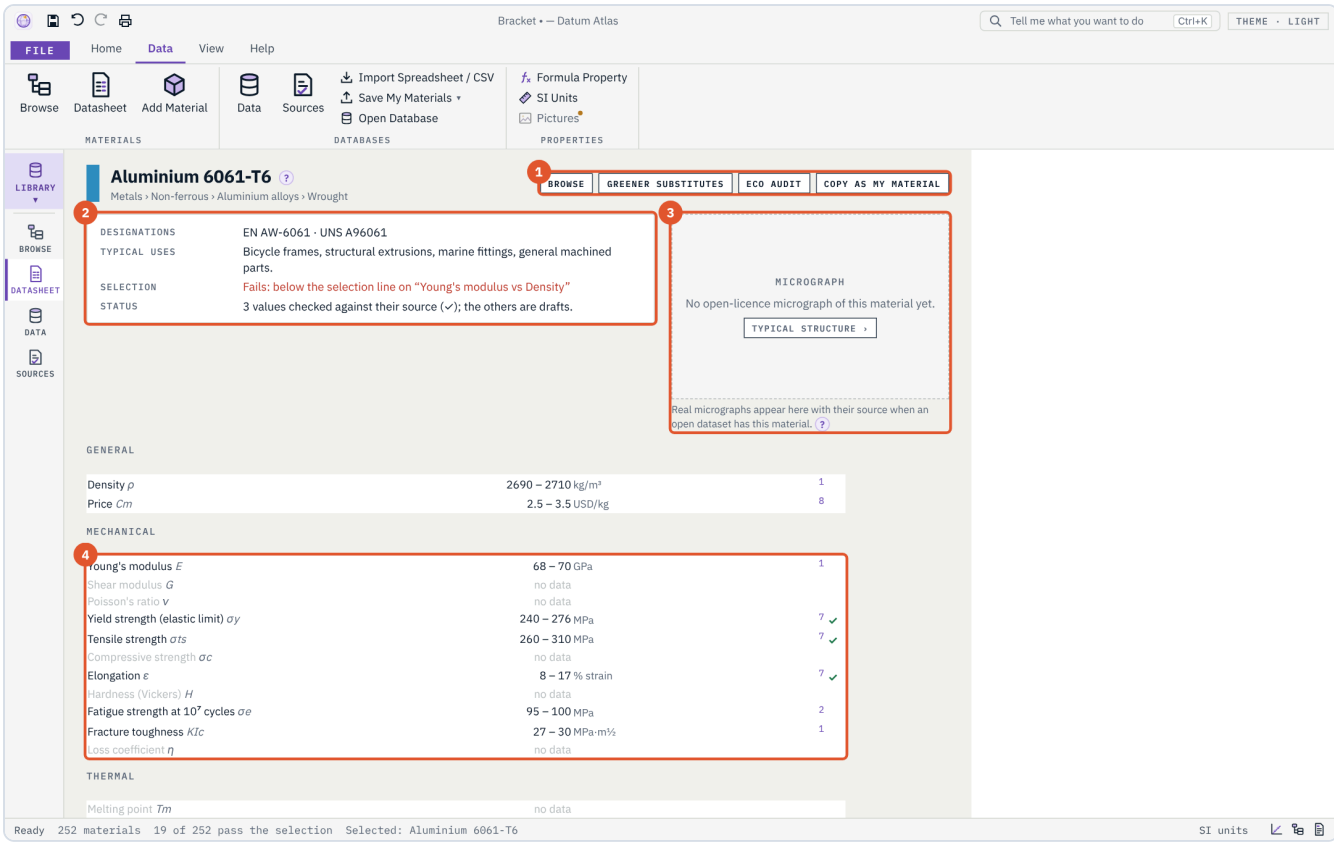
Below them, every material that fails says why: *below the selection line on "Young's modulus vs Density"*, *price above 20 USD/kg*, *no data for max service temperature*. **+ compare** adds a material to Compare. **Select › To a Spreadsheet** saves the whole table (rank, index, × best, the values the selection uses, the reasons) as `.xlsx`; **Report › Results as CSV** as `.csv`.

Browse, datasheets and sources



Browse: the family tree and the materials table.

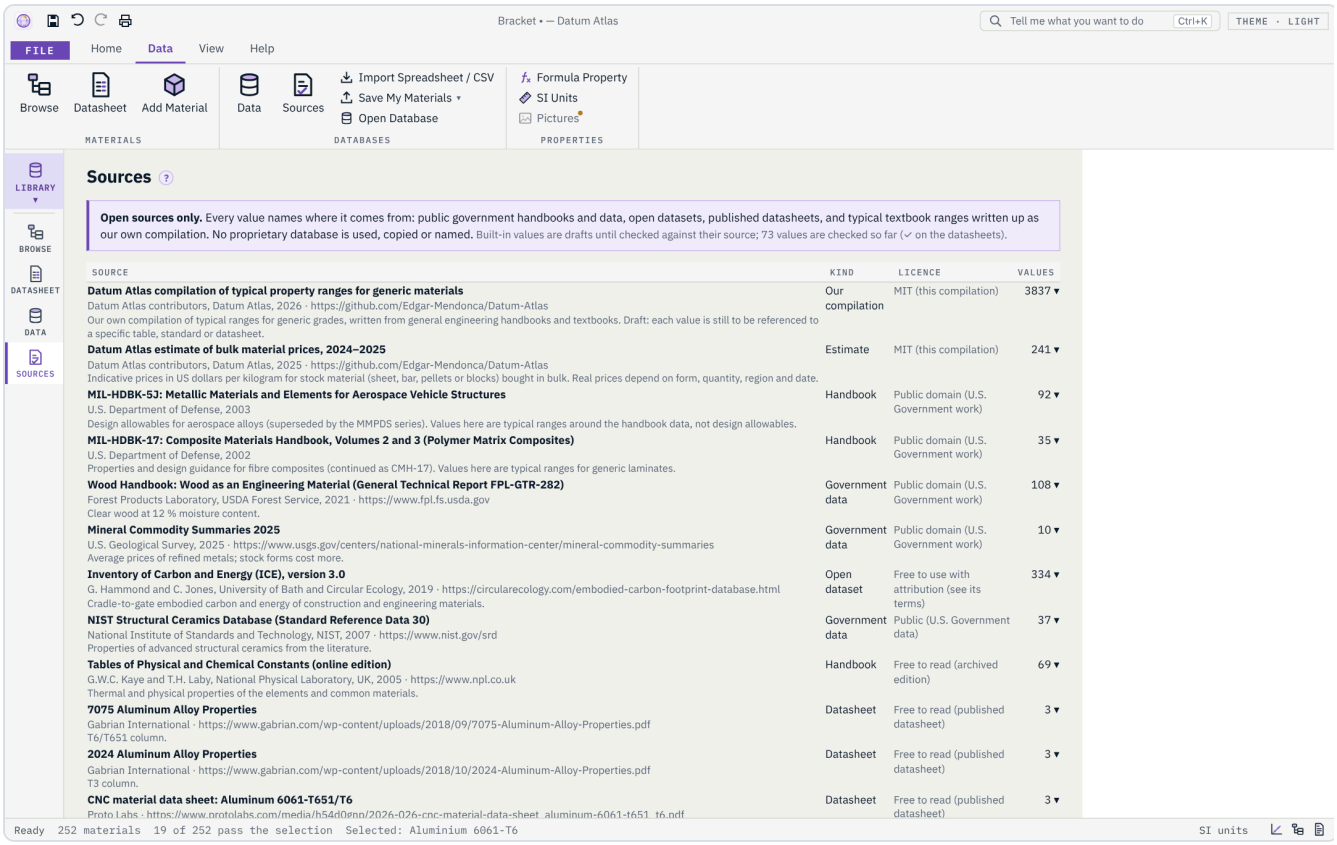
1. **The family tree.** Click a family or class to show only its materials.
2. **Search** by name, designation (UNS, EN, JIS, ASTM), family or use.
3. **The table.** Sort by any heading; choose the columns; select cells, rows or columns and copy them (`Ctrl+C`) into a spreadsheet. `Enter` opens the datasheet.



A datasheet: the facts, every value with its source, and the references.

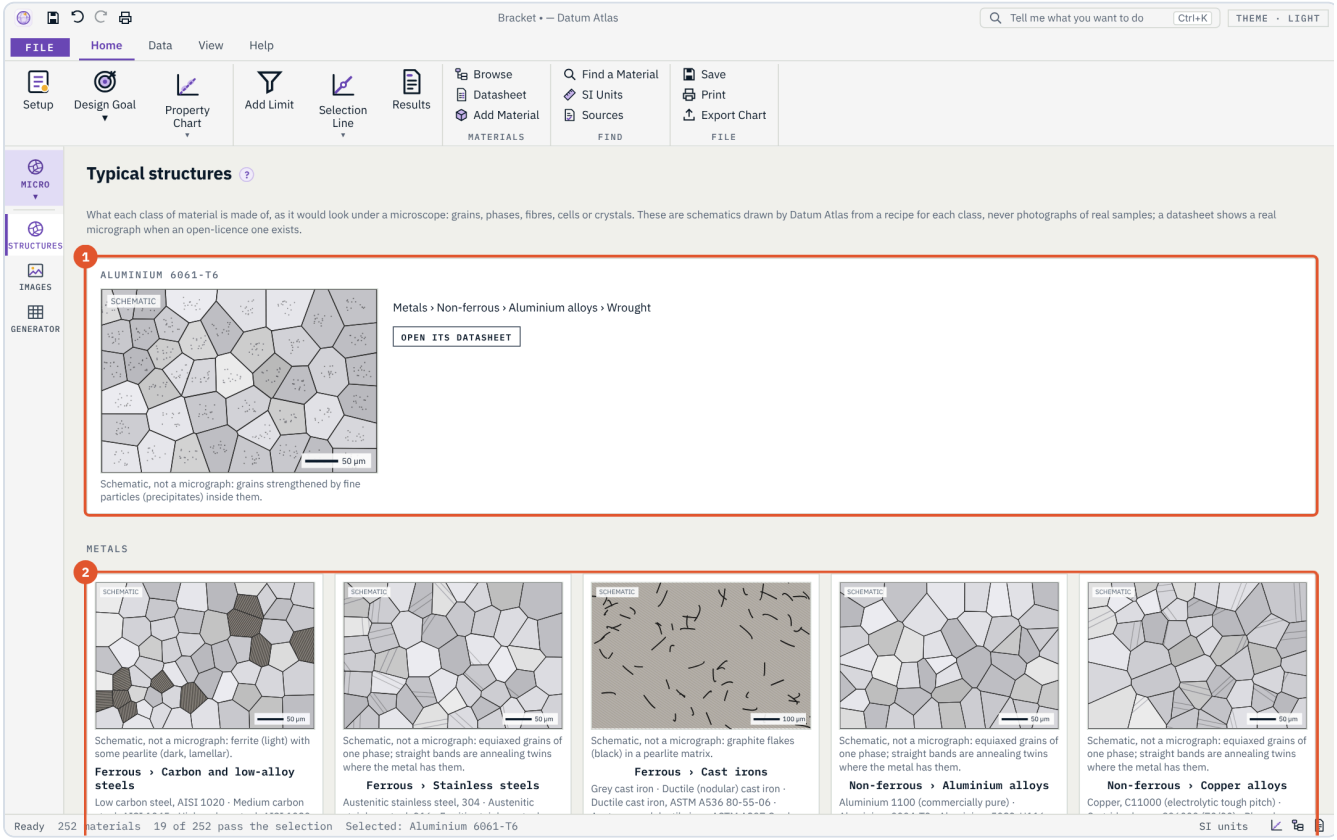
1. **Greener substitutes**, **Eco audit** (adds it to the audit) and **Copy as my material**.
2. **The facts:** designations, other names, typical uses, notes, how it fares in the selection, and its status (draft, or how many values were checked).
3. **The micrograph:** a real micrograph of the material with its source, when an open-licence one exists (none yet: the box says so). **Typical structure** opens the Microstructure module on the material.
4. **The values**, grouped (general, mechanical, thermal, electrical, durability, processing, eco, derived), each with its unit and the number of its reference; ✓ marks a checked value; *no data* is a gap.

The references at the bottom give each source: its title, publisher, year, licence and link. The **Sources** page lists every source, its kind and licence; click one to see which materials and values cite it.



Sources: open and public sources only.

Microstructure: typical structures



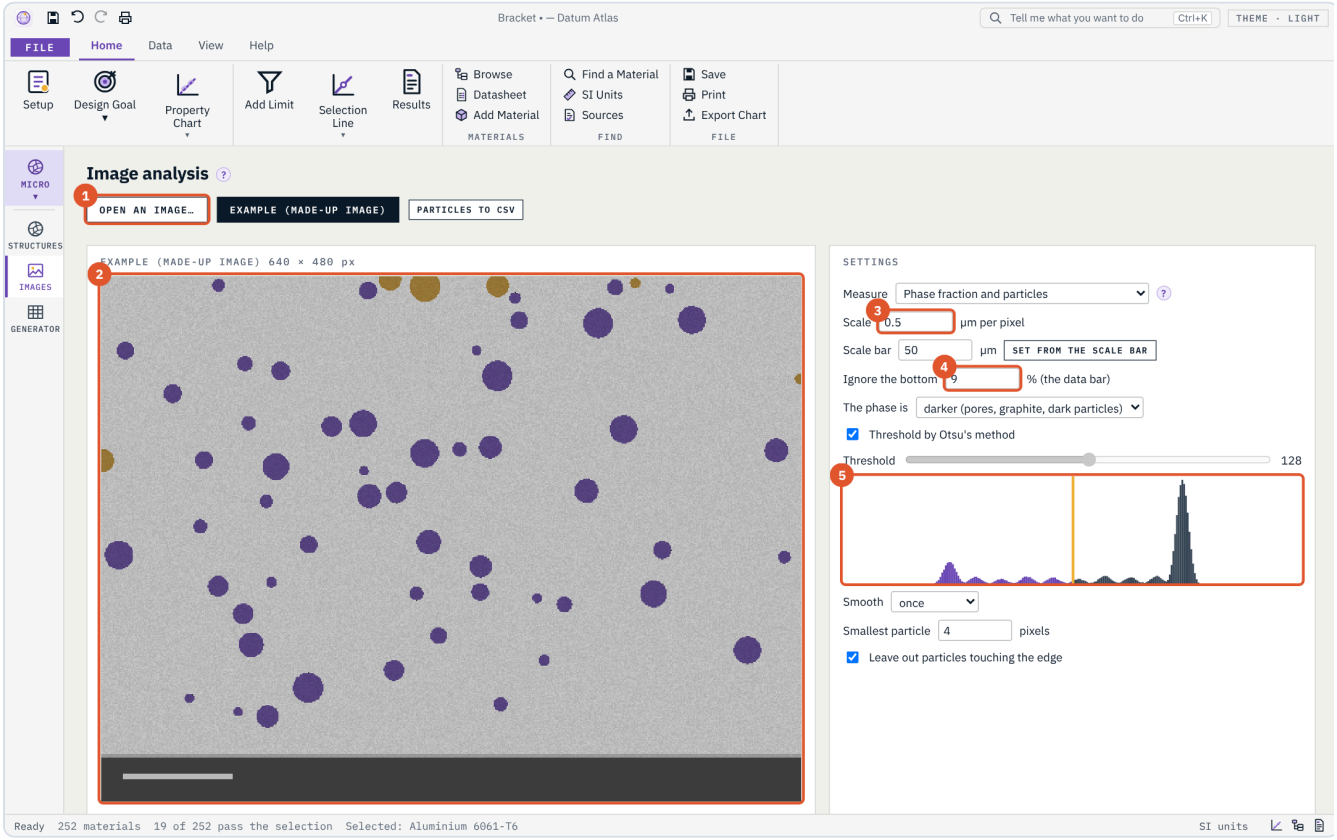
The Microstructure module: a typical structure for every class.

The **Structures** page shows what each class of material is made of, as it would look under a microscope: grains (with twins in copper, nickel and stainless steels), ferrite and pearlite, martensite, graphite in cast irons, dendrites in castings, α + β colonies in titanium alloys, precipitates in age-hardened alloys, sintered ceramics, glass, fibres and plies, foam cells, wood, cork, spherulites in semicrystalline polymers, and more. Each has a scale bar and a caption.

These are **schematics** drawn by Datum Atlas from a recipe for each class, never photographs of real samples. Opened from a datasheet, the page starts with that material's own drawing. The command line draws one too: `datum-atlas-cli microstructure ID`.

Image analysis of your own micrographs, a structure generator, EBSD and DIC will join this module.

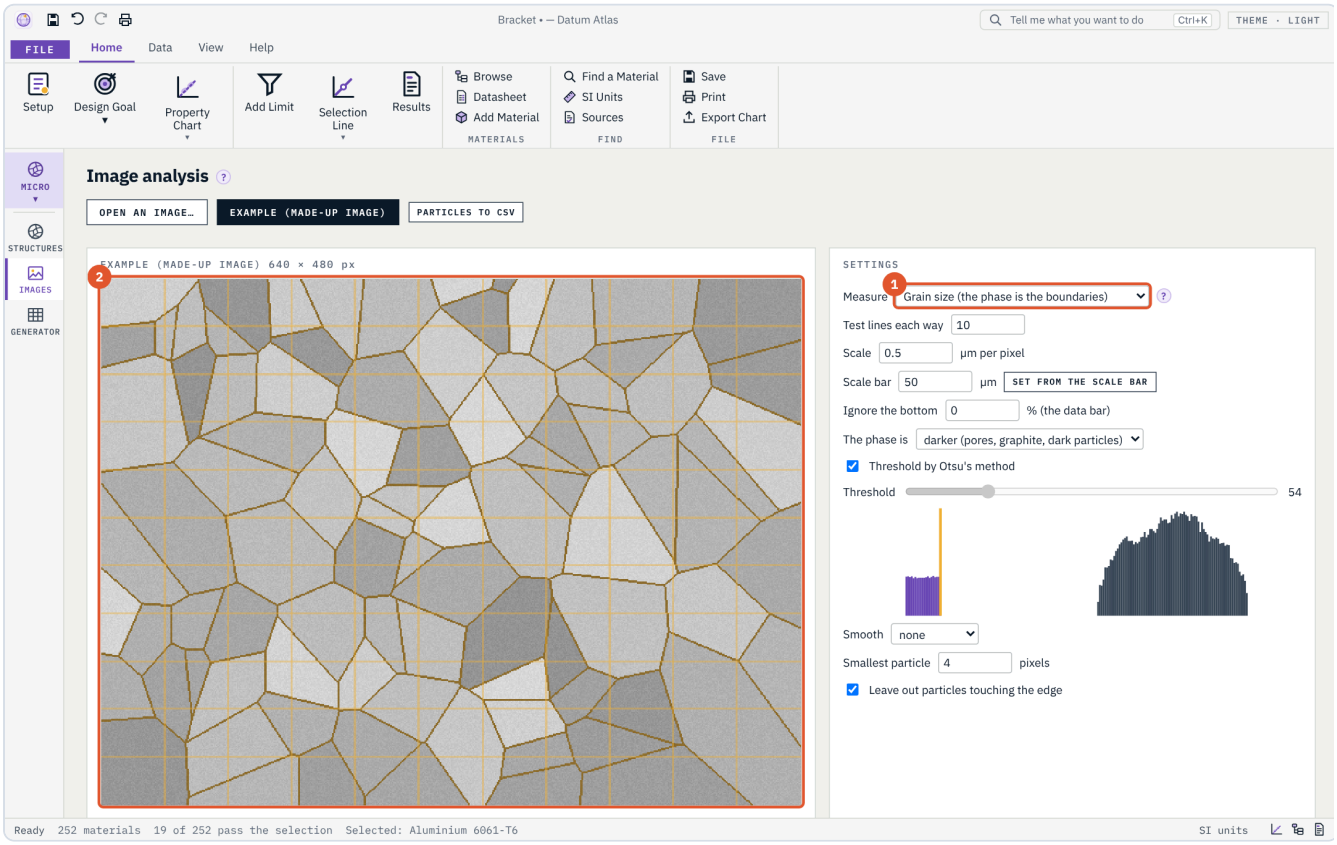
Microstructure: image analysis



Microstructure › Images: the made-up example with its particles measured.

1. **Open an image:** a micrograph as PNG, JPEG, BMP or WebP (save a TIFF as PNG first). It is read in grey; a very large picture is shrunk to about 2.5 million pixels and the page says so. **Example (made-up image)** loads an invented picture to try the page.
2. **The image** with the measurement over it: violet is the phase measured, amber is phase not counted as particles (too small, or touching the edge), grey is left out.
3. **Scale:** type µm per pixel, or click **Set from the scale bar**, type the bar's length and click both of its ends on the image. Without a scale, sizes are in pixels.
4. **Ignore the bottom:** leave out the strip with the microscope's data bar.
5. **Threshold:** say whether the phase is darker (pores, graphite, dark particles) or brighter, and let Otsu's method choose the threshold or set it by hand; the histogram shows the grey levels with the phase in violet and the threshold in amber. Smooth a noisy image, and raise the smallest particle to drop specks.
6. **Results:** the area fraction of the phase (phase fraction or porosity), the number of particles and their number per mm², the equivalent circle diameter (mean, standard deviation, D10, D50, D90), mean area, circularity and aspect ratio.
7. **Size distribution:** a histogram of equivalent diameters and a table of the particles; **Particles to CSV** saves them all.

Particles touching the edge of the region are left out of the sizes, because their whole size is not seen; they still count in the area fraction.

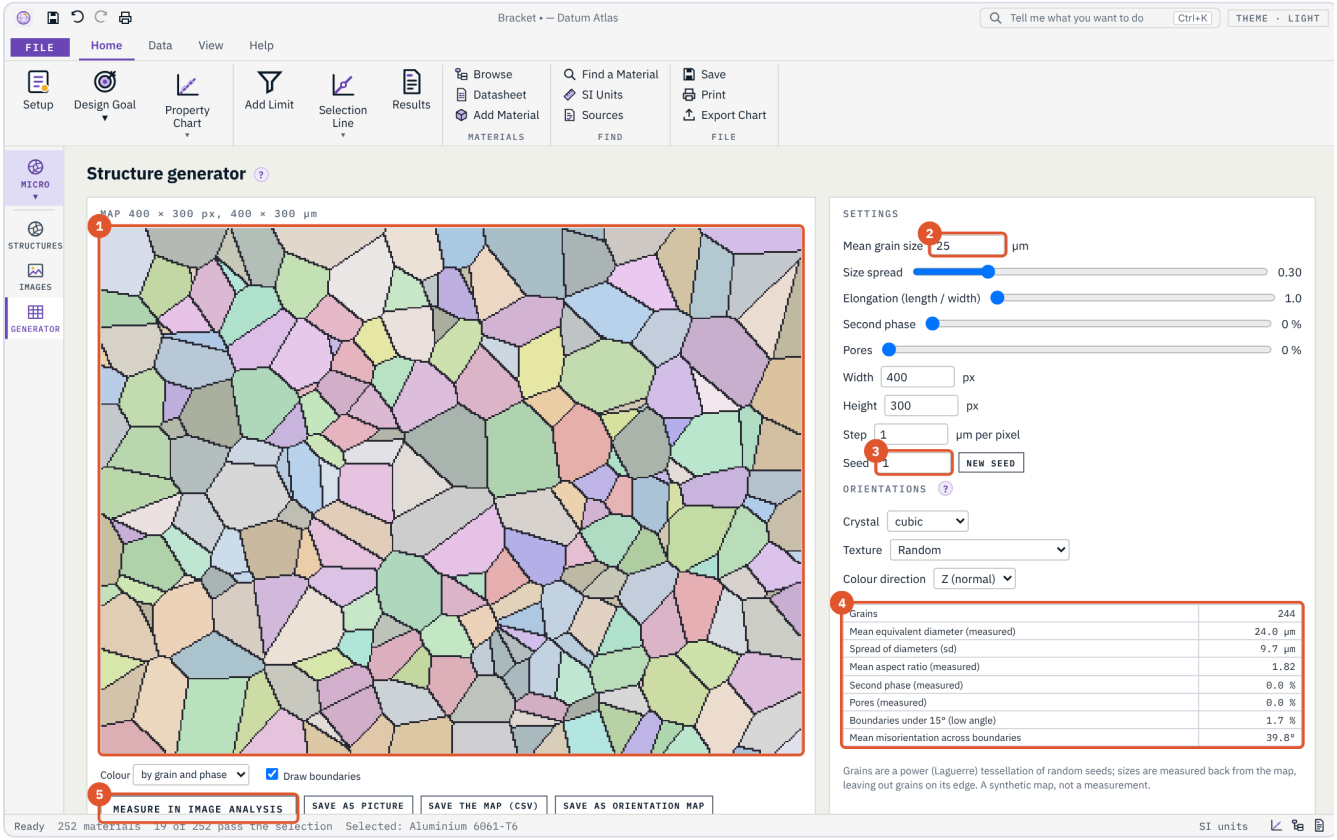


Grain size of the made-up grain image, with the test lines drawn over it.

1. **Measure:** besides phase fraction and particles, choose **Grain size** (the phase is the grain boundaries, usually darker), **Graphite nodularity** (the phase is the graphite) or **Coating thickness** (the phase is the coating, turned to run across the image). Each has a made-up example.
2. **Grain size** lays test lines across the image each way (drawn in amber) and counts the boundaries they cross: the mean lineal intercept is the line length over the count. It also counts the grains, those on the edge as halves, for the mean grain area.
3. **The results:** the mean intercept and grains per mm², with the grain size number G from the standard relations ($G = -6.644 \log_{10} \ell - 3.288$ with ℓ in mm; $G = 3.322 \log_{10} N - 2.954$ with N per mm²).

Nodularity is the share of the graphite in round nodules (at least as round as the limit you set, 0.6 unless you change it), by area and by count. **Coating thickness** is the layer's length in every column of pixels, with its mean, spread, thinnest and thickest, and the profile along the image.

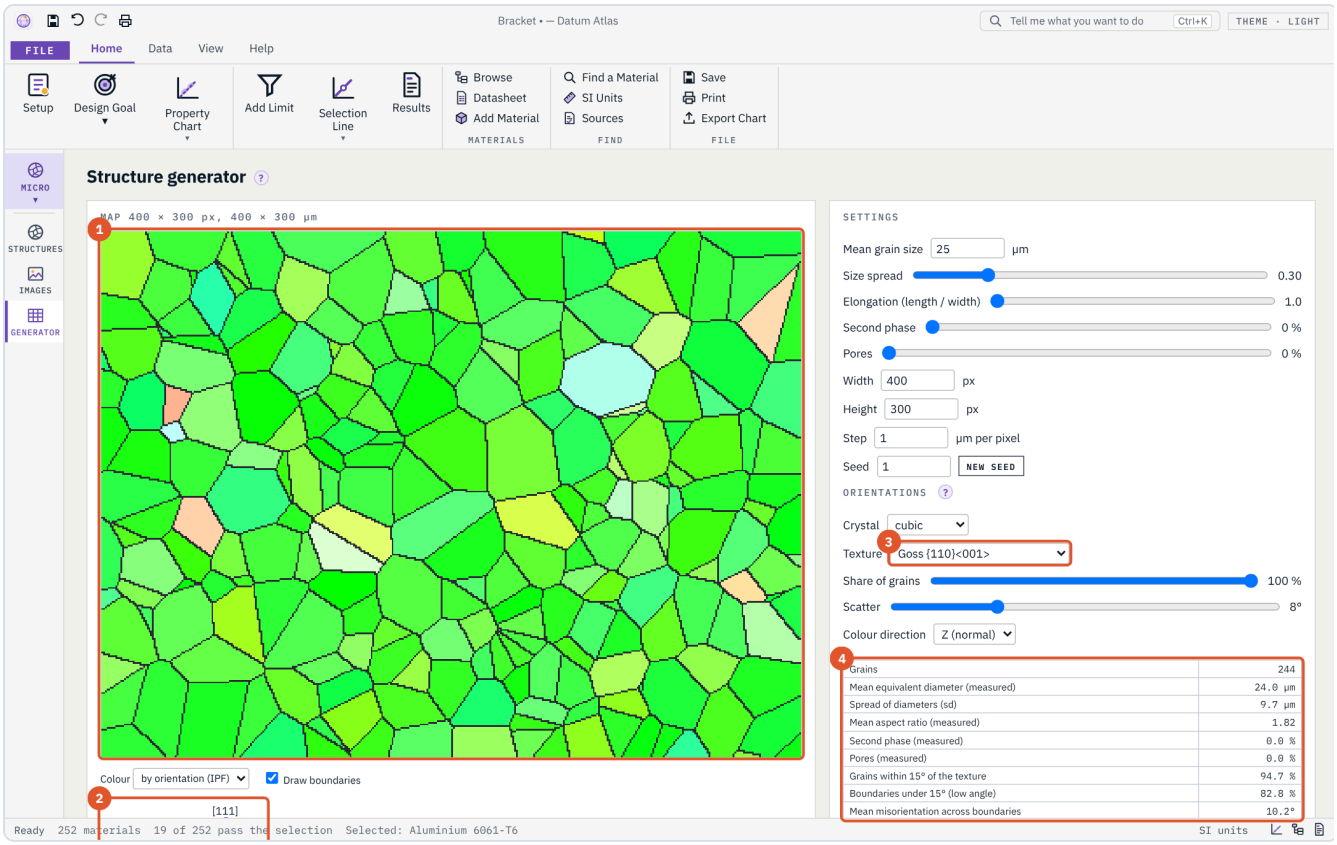
Microstructure: the structure generator



Microstructure › Generator: a synthetic grain map made to order and measured back.

1. **The map:** a synthetic 2D grain structure, one colour per grain; phase 2 is darker and pores are black. Untick **Draw boundaries** to see the grains alone.
2. **Settings:** the mean grain size, the size spread (0 even, 1 very mixed), the elongation (length over width), the second phase and the pores, and the map's size and step. The map redraws as you change them.
3. **Seed:** the same settings and seed always give the same map; **New seed** gives another.
4. **Measured back:** grains, mean equivalent diameter and its spread, aspect ratio, second-phase and pore fractions, from the map itself (grains on its edge left out).
5. **Measure in image analysis** opens the map's grey picture in Images, calibrated, with grain size chosen, so you can try the methods on a structure whose answer you know. **Save as picture**, or save the map as CSV (x, y, grain, phase).

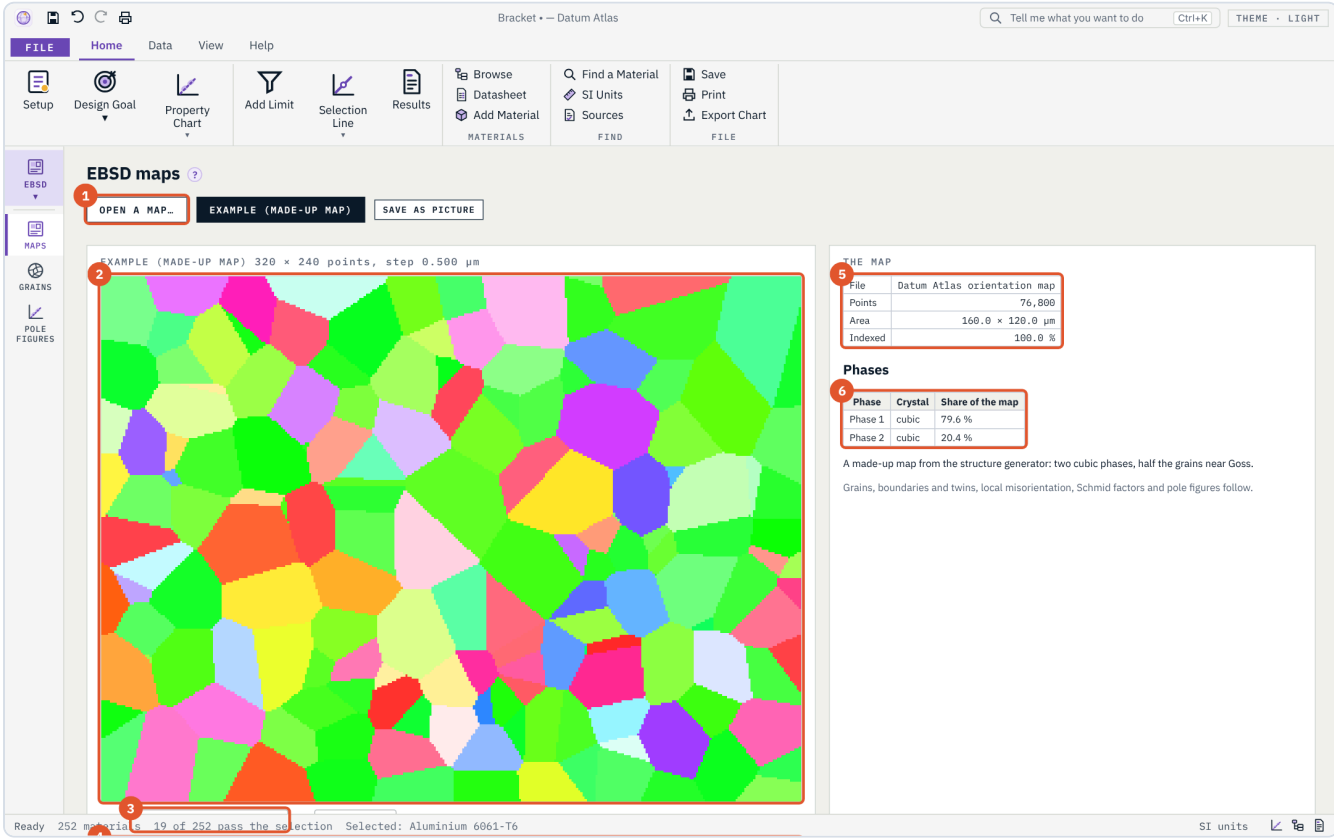
The grains are a power (Laguerre) tessellation of random seeds, the usual model of a random grain structure. It is a synthetic map, not a measurement.



A Goss-textured map coloured by orientation, with the IPF key.

1. **Colour by orientation:** each grain in the inverse pole figure (IPF) colour of the chosen sample direction: the crystal direction lying along it, red at [001], green at [101], blue at [111] for cubic crystals ([0001], [2$\bar{1}\bar{1}$0], [10$\bar{1}$0] for hexagonal).
2. **The key:** the standard triangle with its colours.
3. **Orientations:** the crystal (cubic or hexagonal) and the texture: random, a rolling or annealing component (Cube, Goss, Brass, Copper, S, rotated Cube), a fibre (<111>, <100> or <110> along the normal) or basal for hexagonal metals, with the share of grains that have it and their scatter.
4. **Measured back:** the share of grains within 15° of the texture, the share of low-angle boundaries (under 15°) and the mean misorientation across boundaries (about 40.7° for random cubic grains).
5. **Save as orientation map:** x, y, Euler angles, phase and grain per pixel, as text; the EBSD pages will open it.

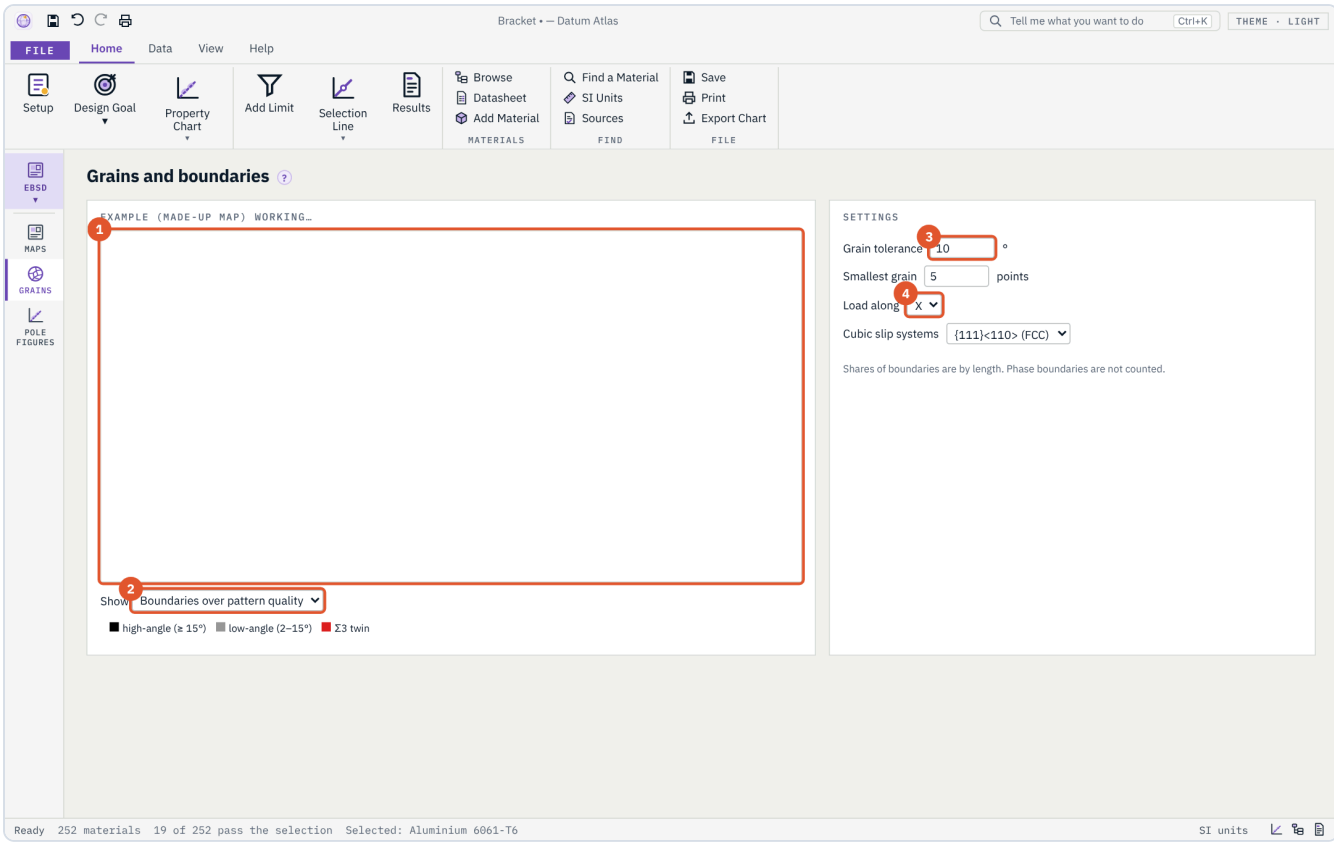
EBSD: orientation maps



EBSD › Maps: the made-up two-phase map coloured by orientation.

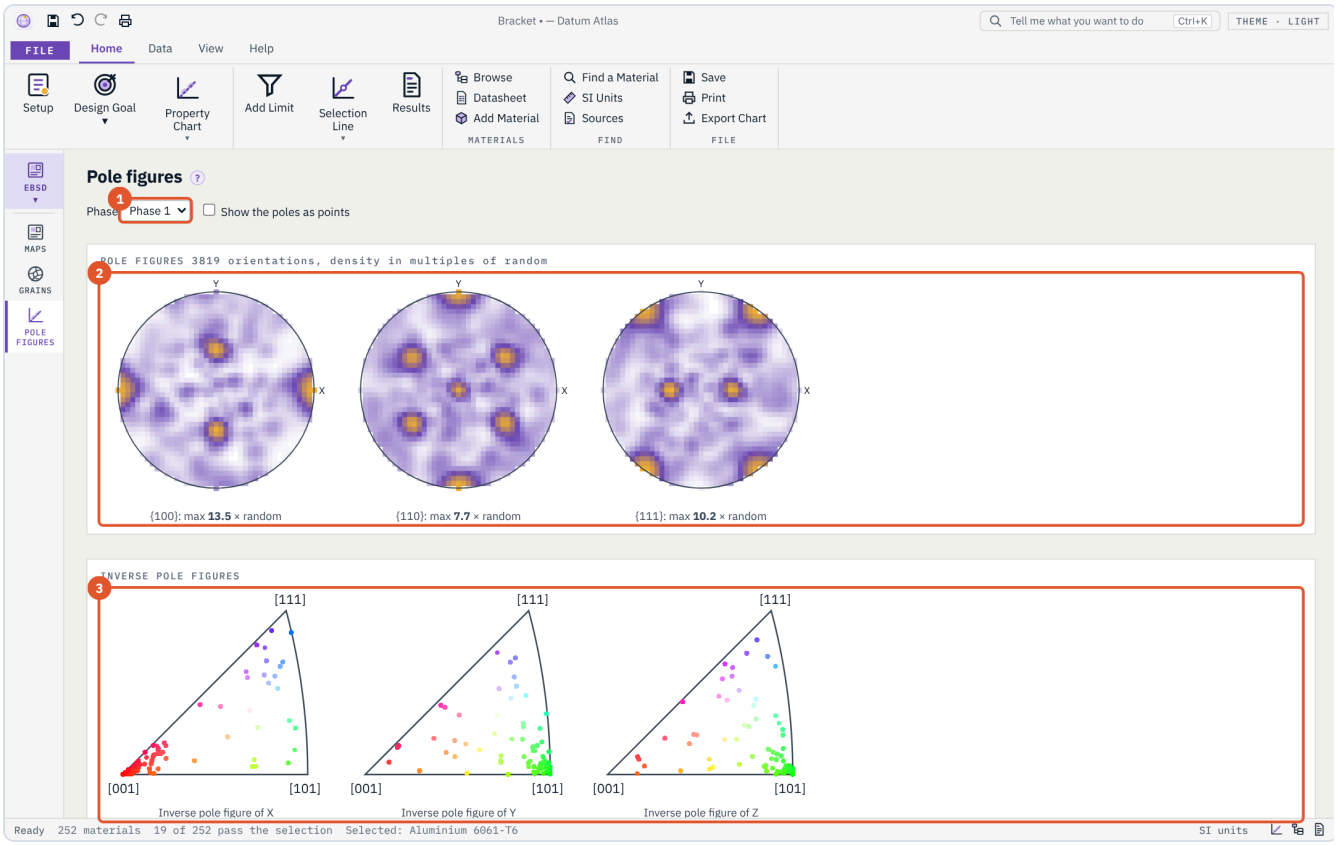
1. **Open a map:** a .ang or .ctf text export from an EBSD system, or an orientation map saved by the structure generator. **Example (made-up map)** opens a two-phase map from the generator.
2. **The map:** one coloured square per point; points that were not indexed are black.
3. **Show:** orientation (the IPF colour of the normal, X or Y: which crystal direction lies along it), phases, or pattern quality and confidence (image quality or band contrast, confidence index or band slope), which show where patterns were weak: boundaries, scratches, dirt.
4. **The key** for each crystal type in the map.
5. **The map:** the file type, the number of points, the area, and the share of points indexed.
6. **Phases:** each phase with its crystal type and its share of the map.

A map on a hexagonal grid is shown on a square grid of the same step.



EBSD › Grains: boundaries over pattern quality, with the grain statistics.

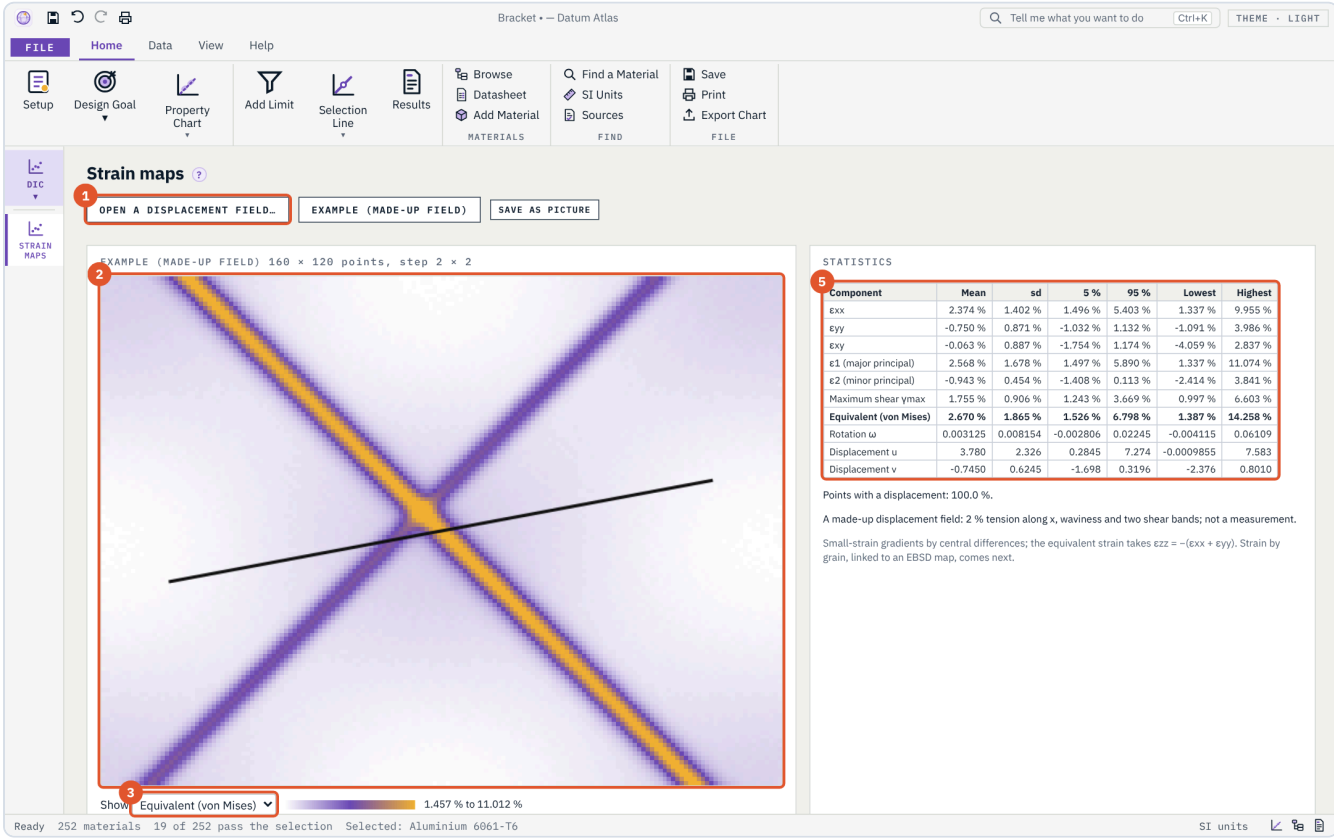
1. **The map:** boundaries over the pattern quality (high-angle black, low-angle grey, Σ3 twins red), or the grains, the local misorientation (KAM) or the Schmid factor.
2. **Show:** which of these to draw.
3. **Grain tolerance:** neighbouring points of the same phase closer than this belong to one grain (10° is usual); grains smaller than the smallest size are left out.
4. **Load along** X, Y or Z for the Schmid factor, with FCC or BCC slip systems for cubic phases (hexagonal phases use basal and prismatic <a> slip).
5. **Results:** grains, their mean equivalent diameter, spread and area (grains on the edge left out), the shares of high-angle, low-angle and Σ3 twin boundaries by length, the boundary length per area, the mean KAM and the mean Schmid factor.



EBSD › Pole figures: the made-up map's half-Goss texture.

1. **Phase:** the phase to plot.
2. **Pole figures:** where the normals of the {100}, {110} and {111} planes (cubic; {0001}, {10$\bar{1}$0} and {11$\bar{2}$0} for hexagonal) point in the sample, seen down Z with X to the right and Y up. The shading is the density in multiples of a random distribution (1 everywhere for no texture); the caption gives the strongest peak. **Show the poles as points** to see single grains.
3. **Inverse pole figures** of X, Y and Z: which crystal directions lie along each, in the standard triangle and their IPF colours.

DIC: strain maps

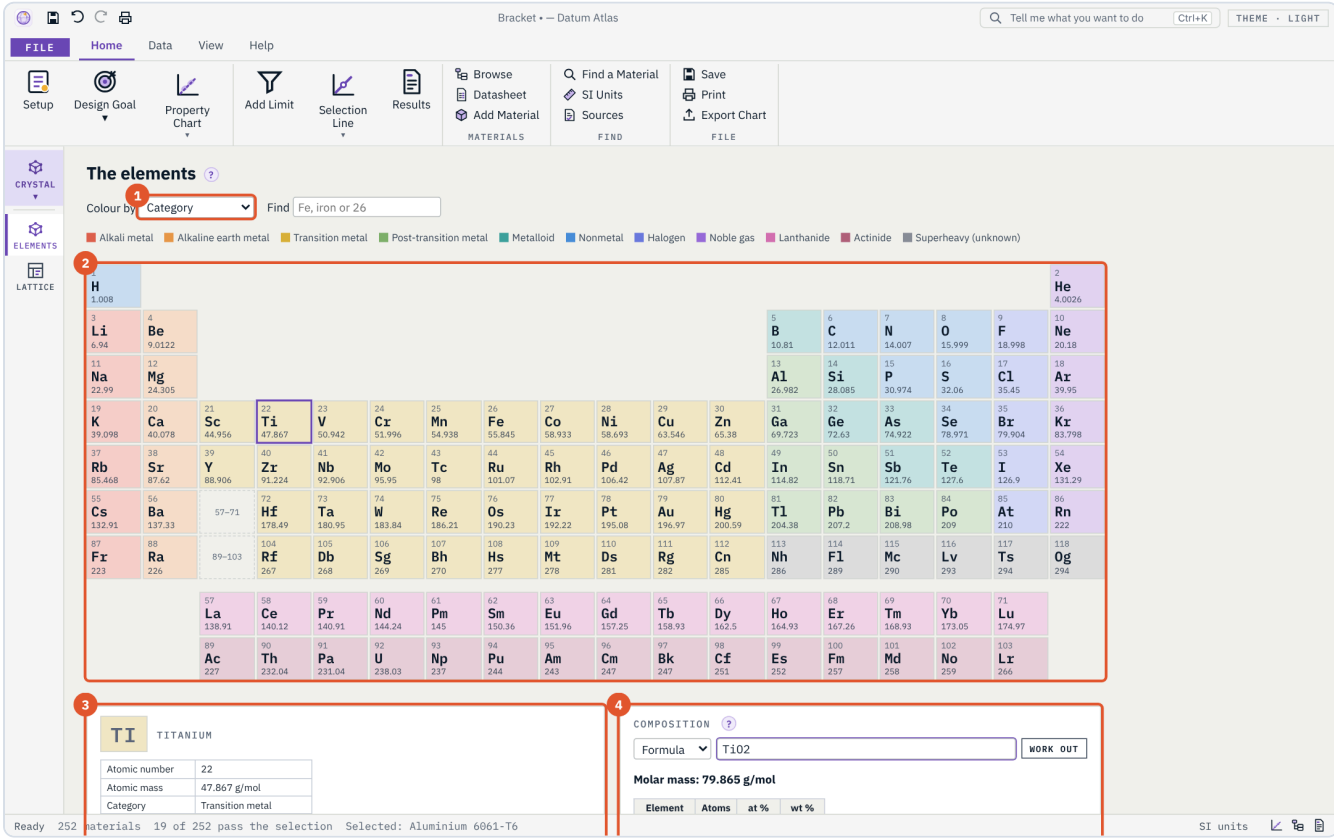


DIC › Strain maps: the made-up field's equivalent strain, with a profile across the shear bands.

1. **Open a displacement field** exported by your DIC software: a CSV, text or spreadsheet file with columns x, y, u and v in one length unit. Headings such as x, y, u, v or ux, uy are found; without headings the first four columns are taken in that order. **Example (made-up field)** opens 2 % tension with two shear bands.
2. **The map** of the chosen component, with grey where there is no displacement. Click two points for a profile along the line between them.
3. **Show:** εxx, εyy, εxy, the principal strains ε1 and ε2, the maximum shear, the von Mises equivalent strain, the rotation, or the displacements. Signed components use a blue–white–red scale centred on zero.
4. **The profile** of the component along the line.
5. **Statistics** of every component: mean, standard deviation, 5 % and 95 % values, lowest and highest.

Strains are small-strain gradients by central differences on the grid; the equivalent strain takes εzz = −(εxx + εyy). Strain by grain, linked to an EBSD map, comes next.

Crystal: the elements

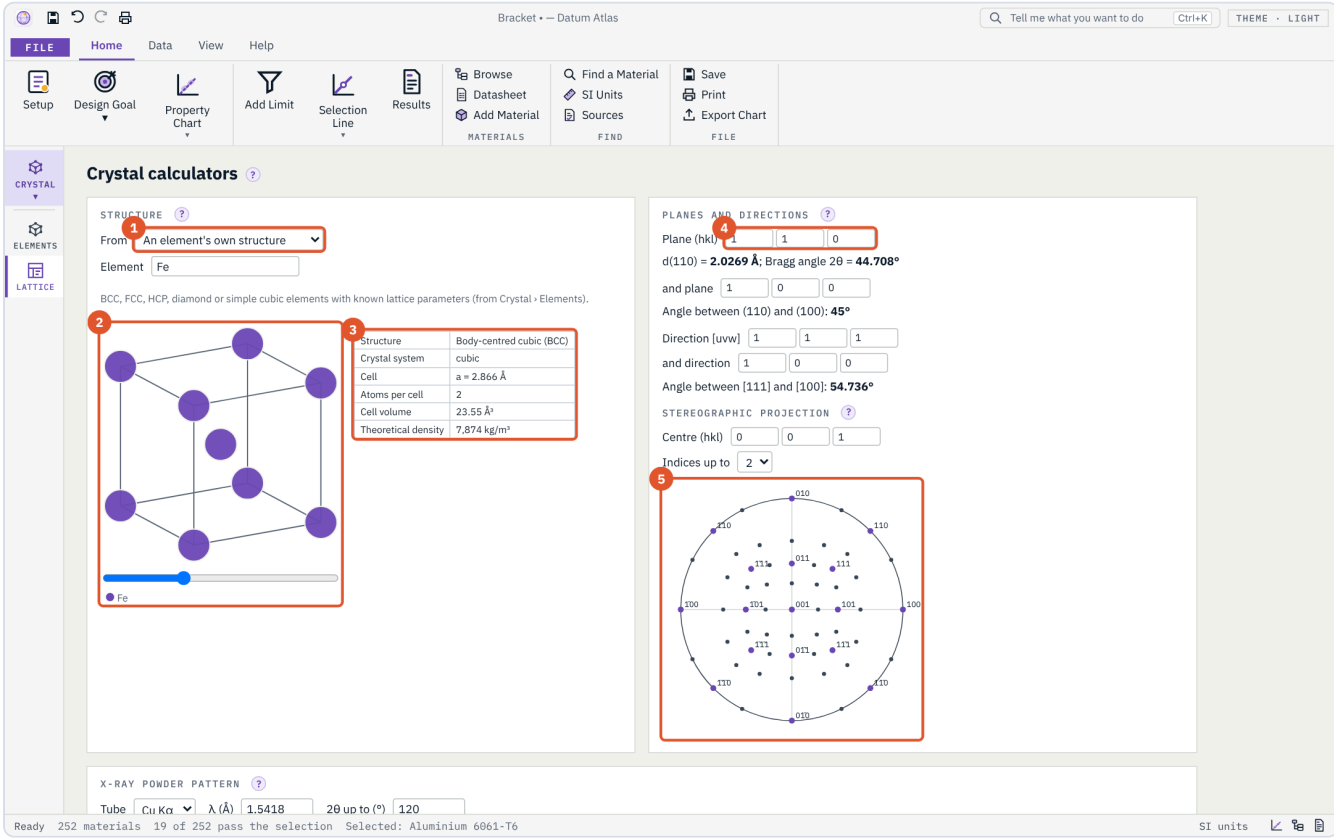


Crystal › Elements: the periodic table, an element card and the composition box.

1. **Colour by:** colour the table by category, or by density, melting point, electronegativity or atomic mass to see how a property runs across it (pale is low, deep is high).
2. **The table:** click an element, or type its symbol, name or number in **Find** and press Enter.
3. **The element card:** atomic mass, category, group and period, state, density, melting and boiling points, crystal structure, lattice parameters and electronegativity, with their sources. A value nobody knows reliably, as for most of the heaviest elements, is left blank and the table hatches it; it is never shown as zero. **In the library** lists the materials built on the element.
4. **Composition:** the molar mass and each element's atom and weight percent of a formula (Fe2O3, Al2(SO4)3, CuSO4·5H2O), or the change between weight and atom percent for an alloy: write each element and its percent, and leave one without a number as the balance (Al Cu 4.4 Mg 1.5 Mn 0.6).

The command line does the same: `datum-atlas-cli element Fe` and `datum-atlas-cli composition --wt Fe C 0.8`. **Its crystal, X-ray peaks and projection** on the card of a metal or semiconductor opens it in the Lattice page.

Crystal: lattice calculators



Crystal › Lattice: a structure, its planes and directions, and a stereographic projection.

1. **From:** an element's own structure (BCC iron, FCC aluminium, HCP magnesium, diamond silicon…), a structure type (simple cubic, BCC, FCC, HCP, diamond, rock salt, caesium chloride, zinc blende, fluorite) with its lattice parameter and elements, or **Any cell** with all six lattice parameters for tetragonal, orthorhombic, monoclinic and triclinic crystals.
2. **The cell:** every atom of the unit cell with its copies on the faces, edges and corners. The slider turns it.
3. **The facts:** crystal system, cell, atoms per cell, cell volume and the theoretical density (the mass of the atoms in the cell over its volume).
4. **Planes and directions:** type Miller indices. The page gives the spacing d of plane (hkl) and its Bragg angle, the angle between two planes and between two directions, and for hexagonal cells the four-index forms (hkil) and [UVTW]. Negative indices show with a bar.
5. **Stereographic projection:** the plane normals seen down the normal of the centre plane, up to indices 1, 2 or 3; the {100}, {110} and {111} poles are labelled.

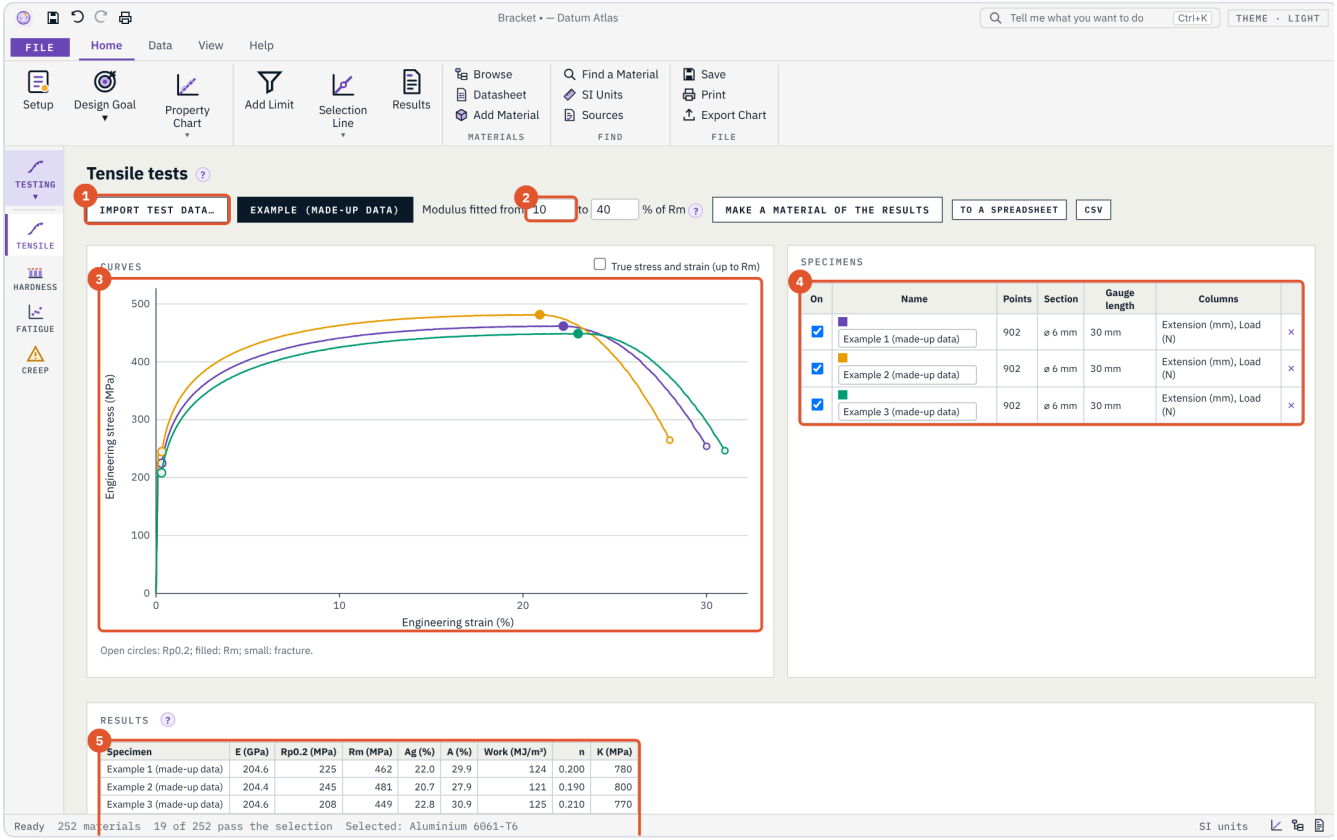


The X-ray powder pattern of BCC iron with Cu Kα radiation.

1. **Tube:** Cu, Co, Mo, Cr, Fe or Ag Kα, or type any wavelength.
2. **The pattern:** a line for each family of planes at the 2θ of Bragg's law, as high as its rough intensity. Families the structure cancels, such as (100) in BCC or (200) in diamond, are left out.
3. **The table:** indices, multiplicity, spacing, 2θ and intensity.

The positions are exact for the cell; the intensities are approximate (each atom scatters in proportion to its atomic number, without the fall-off with angle), so they show which lines are there and roughly how strong. On the command line: `datum-atlas-cli crystal Fe`, or `--ask` with any structure.

Testing: tensile tests

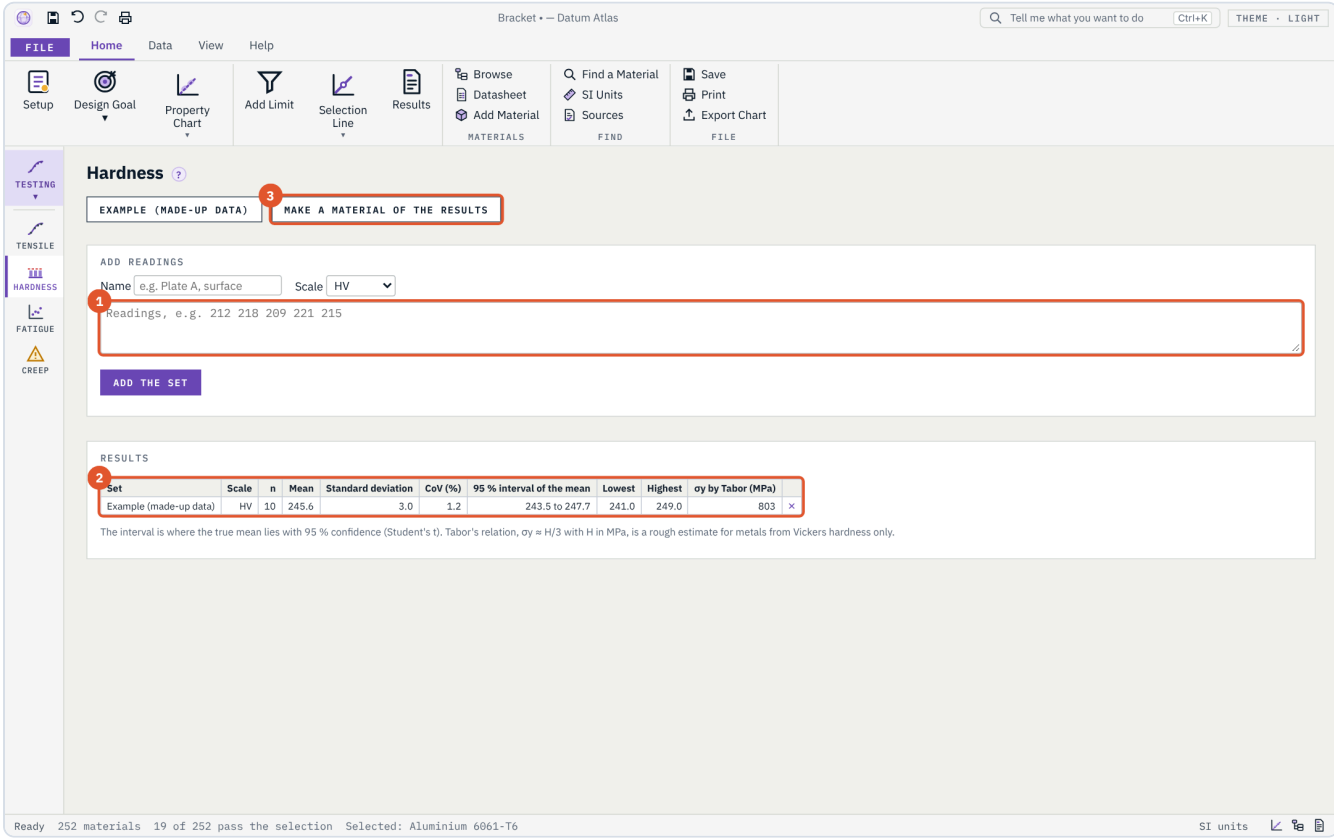


Testing › Tensile: three specimens of the made-up example, their curves and results.

1. **Import test data:** a test machine's export as CSV, text or a spreadsheet, with a column of extension or strain and one of load or stress. Lines of heading at the top are skipped and the columns are guessed from their headings; check them, say what they hold (extension in mm, strain, or strain in %; load in N or kN, or stress in MPa), give the section (round, rectangular or an area) and the gauge length, and **Add the specimen**. **Example (made-up data)** adds three invented steel-like specimens to try the page; they are not a measurement of any material.
2. **Modulus window:** Young's modulus is fitted as a straight line to the loading part of the curve between two stresses, as percentages of the tensile strength (10 % to 40 % unless you change it). Where the line meets zero stress sets the strain origin, so slack at the start does not count.
3. **Curves:** engineering stress against strain for every specimen that is on, with Rp0.2 (open circle), Rm (filled) and fracture (small); or true stress against true strain up to Rm.
4. **Specimens:** rename, untick to leave one out of the statistics (a bad grip, a break outside the gauge length), or remove.
5. **Results:** for each specimen E, Rp0.2, Rm, Ag (plastic strain at Rm), A (plastic strain at fracture), the work to fracture, and n and K of $\sigma = K\,\varepsilon^{n}$ fitted to true stress and true plastic strain between 0.2 % plastic strain and Ag; then the mean, standard deviation, lowest and highest. A value that cannot be worked out is blank, with a note saying why.

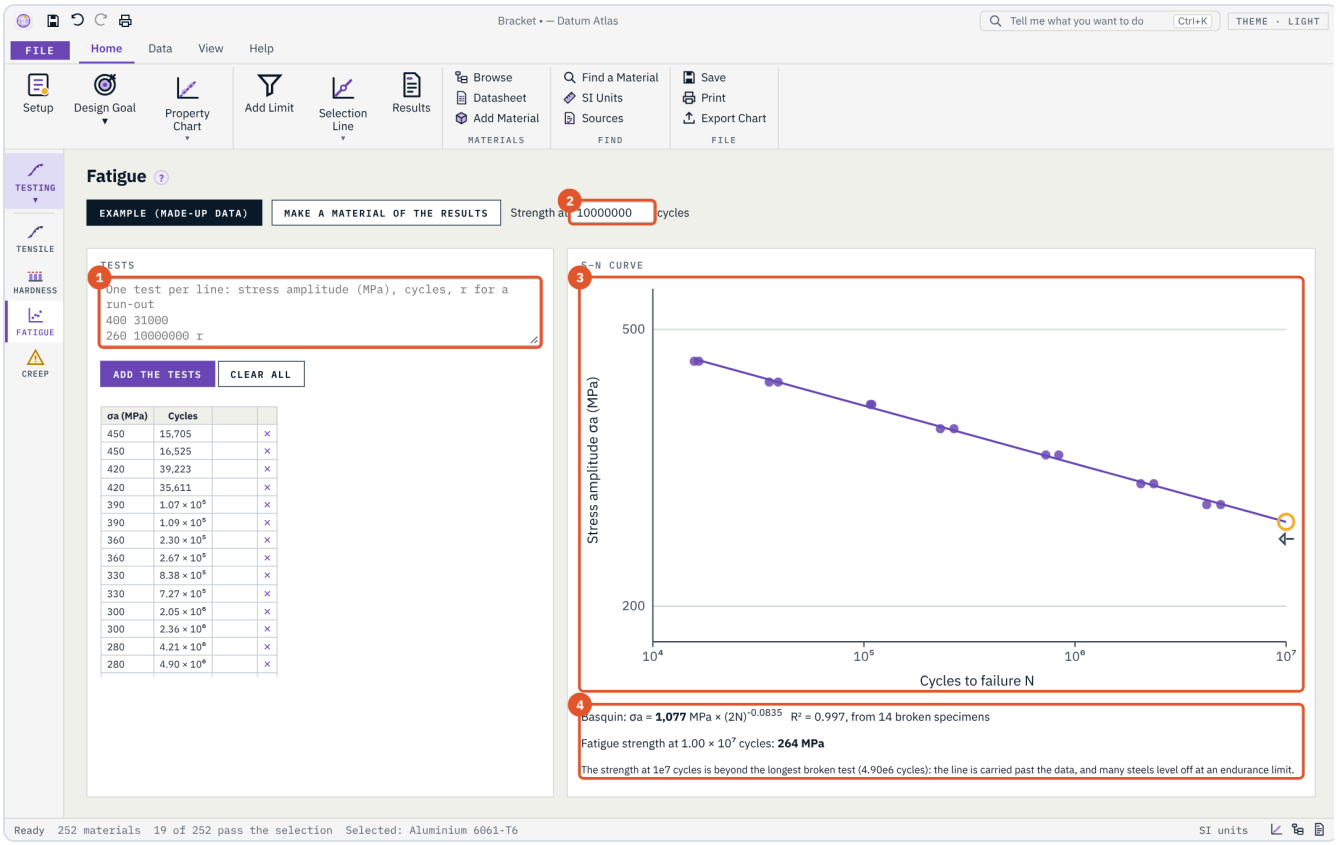
A modulus from crosshead extension comes out low because the machine stretches too: use an extensometer for E. The specimens are saved in the project; **To a spreadsheet** and **CSV** export the results, and `datum-atlas-cli tensile FILE` prints them.

Testing: hardness, fatigue and creep



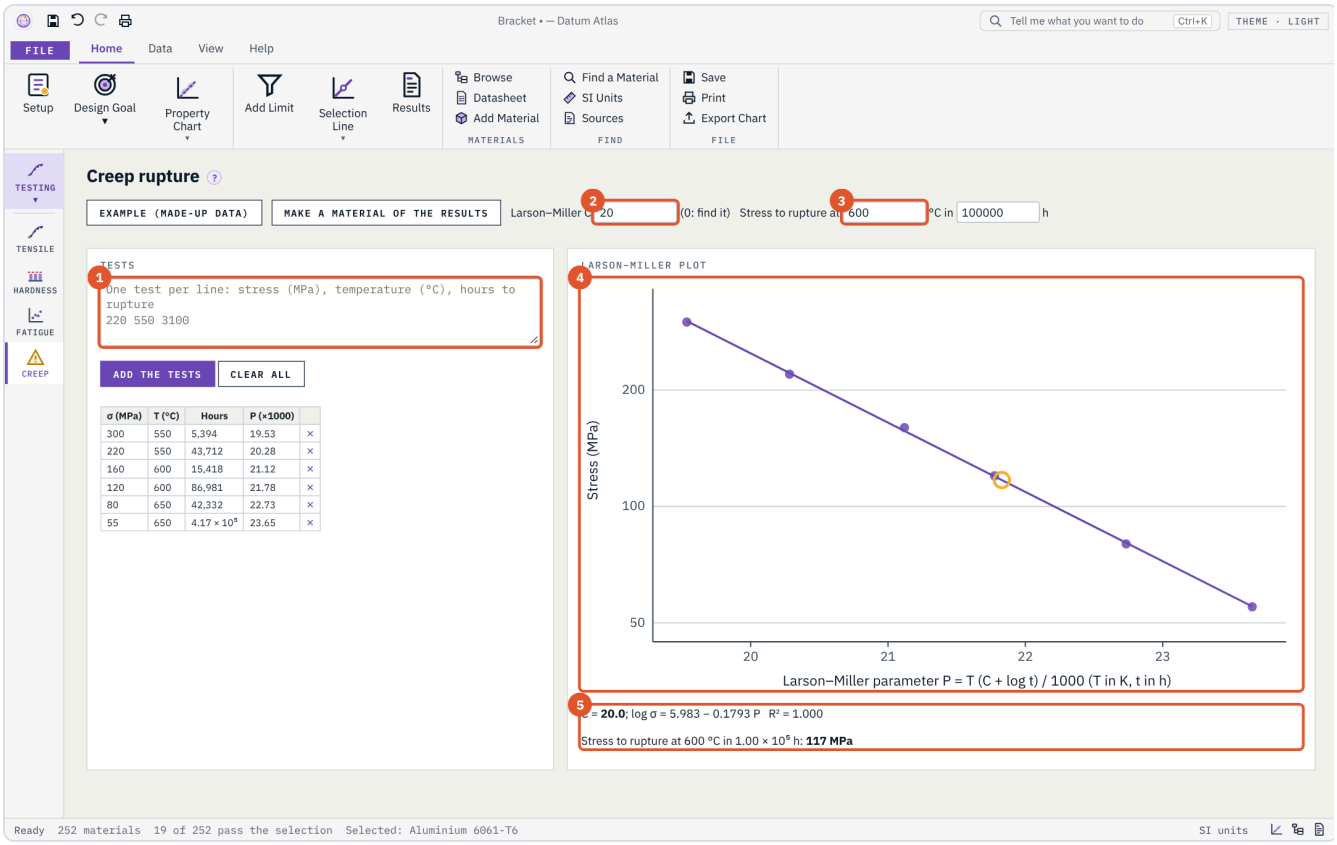
Testing › Hardness: readings, their statistics and Make a material of the results.

1. **Readings:** name the set, pick the scale (HV, HB, HK, HRC, HRB, HR15N, Shore A or D) and paste or type the readings, then **Add the set**.
2. **Results:** the mean, standard deviation, coefficient of variation, the 95 % interval of the mean (Student's t), lowest and highest; for Vickers hardness, a rough yield strength by Tabor's relation, σy ≈ H/3. Scales are not converted into one another.
3. **Make a material of the results:** a material of your own from every result in the project (modulus, Rp0.2, tensile strength and elongation from the tensile specimens that are on, Vickers hardness, the fatigue strength at 10^7 cycles), each value the range over the specimens. The Add Material dialog opens filled in, with this project's tests as the source; add the density and Save.



Testing › Fatigue: an S–N curve fitted with Basquin's law.

1. **Tests:** one per line: stress amplitude in MPa, cycles, and r for a run-out (a specimen that did not break).
2. **Strength at:** the cycles for the fatigue strength (10^7 unless you change it).
3. **S–N curve:** the tests on log scales, run-outs as arrows, and the line of Basquin's law σa = σf′ (2N)^b fitted to the broken specimens; the amber ring is the strength asked for.
4. **The fit:** σf′, b, R^2, and the strength. When the strength lies beyond the longest broken test the page says so: many steels level off at an endurance limit.

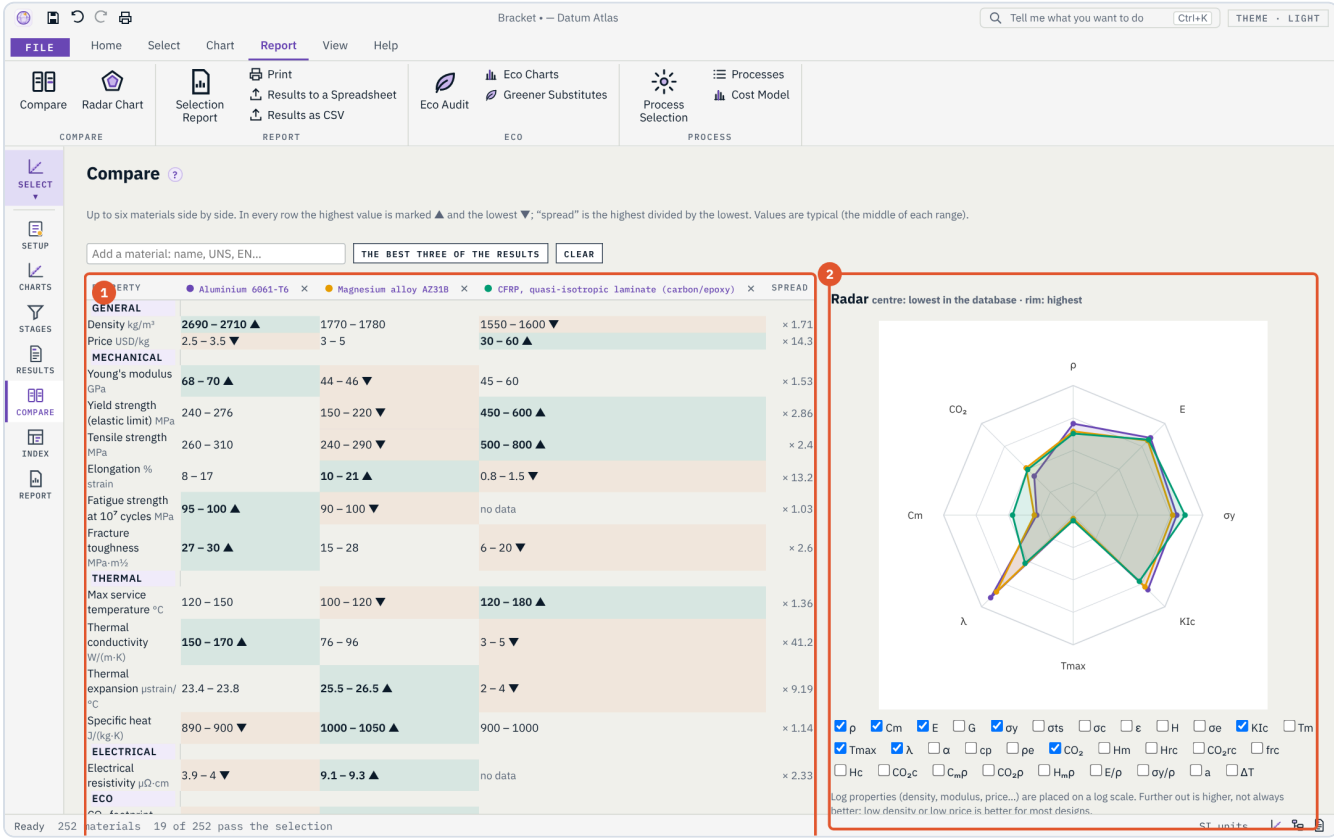


Testing › Creep: rupture tests on a Larson–Miller plot.

1. **Tests:** one per line: stress in MPa, temperature in °C, hours to rupture.
2. **Larson–Miller C:** the constant in P = T (C + log10 t), with T in kelvin and t in hours; often near 20. Type 0 and the page finds the C that fits your tests best.
3. **The question:** the stress to rupture at a temperature and a time.
4. **The plot:** stress (log scale) against P, with the fitted line; the amber ring is the answer.
5. **The fit:** C, the line, R² and the stress to rupture, with a warning when the question lies outside your tests.

Example (made-up data) on these pages fills them with invented data to try them. `datum-atlas-cli lab FILE` prints the results.

Compare

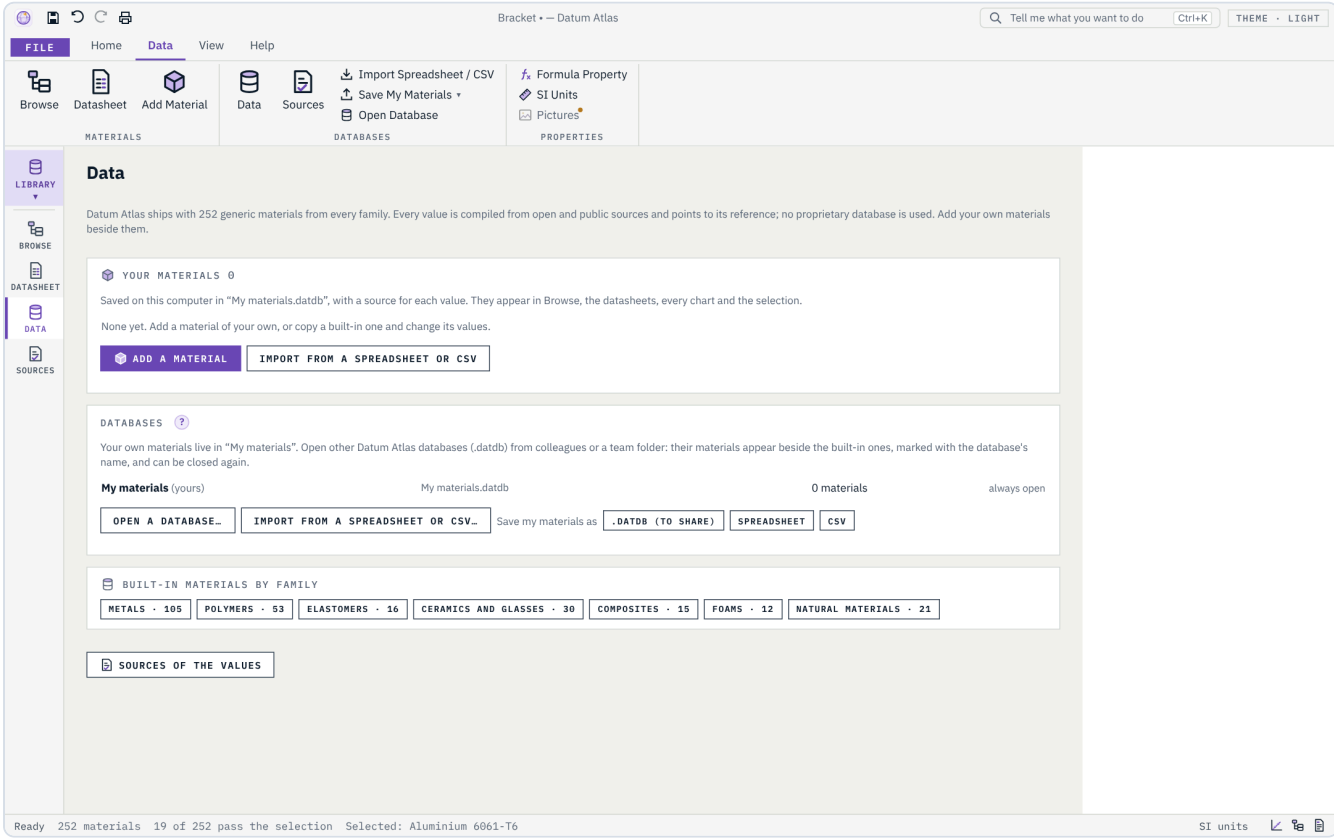


Compare: three materials side by side, and the radar chart.

1. **The table.** Up to six materials, property by property: the highest of each row is marked ▲ and the lowest ▼, and **spread** is the highest divided by the lowest.
2. **The radar chart.** Each material's outline, each property placed between the lowest (centre) and the highest (rim) value in the whole database. Tick the properties to show. Further out is higher, not always better: low density or low price is better for most designs.

Add materials with the box at the top, from Results (**+ compare**), from the card beside the chart, or **The best three of the results**.

Your own materials and databases

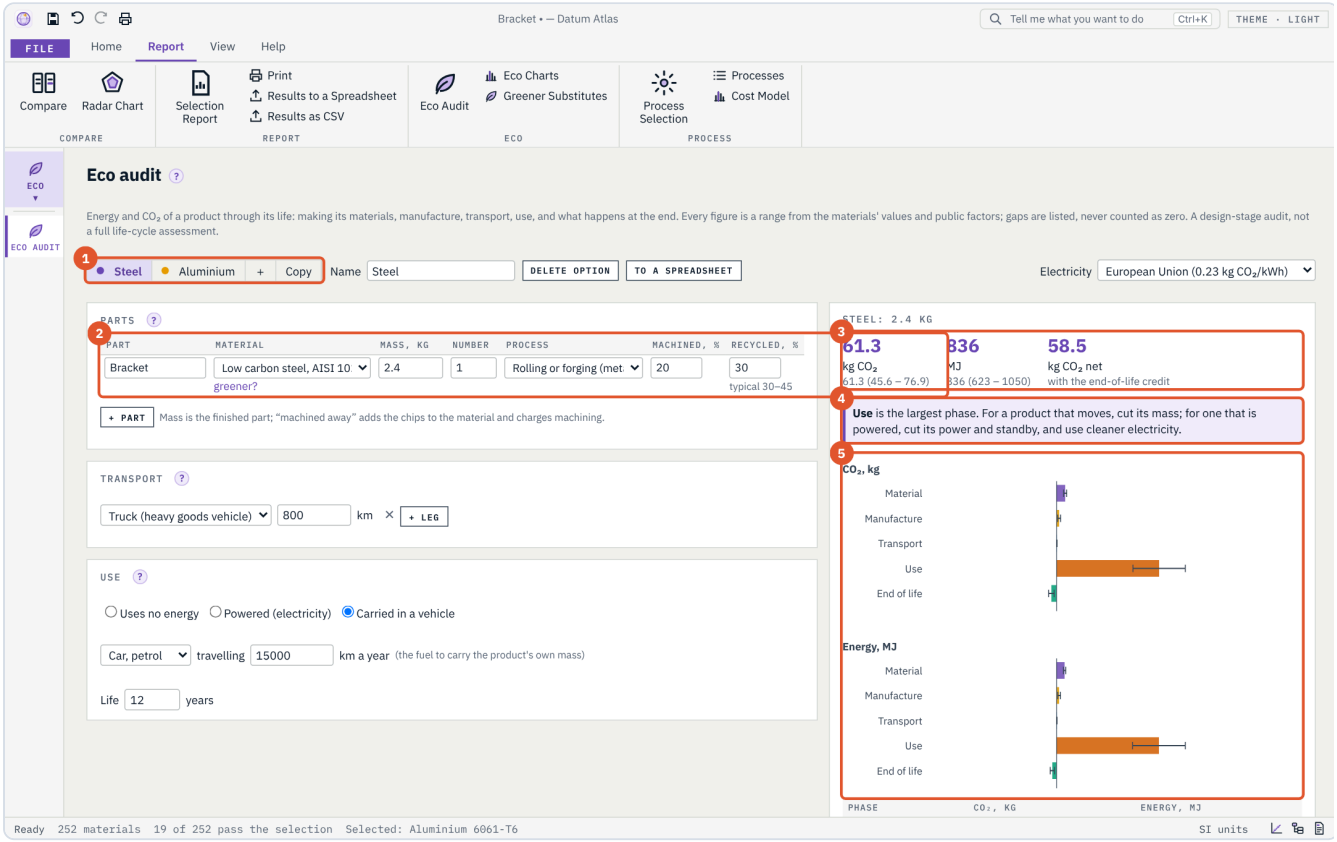


Data: your materials, databases, families and properties.

- **Add Material** opens a form that says what is needed: a name, the family, density, at least one more property, the low and high of each value in the unit shown, and where the values come from (a book, a datasheet, a test report or your own measurement).
- **Import Spreadsheet / CSV** reads a table: each column is matched to a property by its heading (change any that are wrong); a cell may hold one value or a range (2690 – 2710); headings in imperial units are converted. You see the rows before they are added, and values that cannot be used are listed, never added as zero.
- **Open Database** opens another `.datdb` (a colleague's, a team folder's) beside the built-in data; its materials are marked with its name and can be closed again.
- **Save My Materials** writes your materials as a `.datdb` to share, a spreadsheet or CSV.

The eco audit

The eco audit adds up the energy and CO_2 of a product through its life: making its materials, manufacture, transport, use and the end of its life. It shows which phase matters most, so you know where a change helps.



The eco audit of a bracket: parts, totals, the largest phase and the phase bars.

1. **Design options.** Up to four versions of the product; **Copy** one to change one thing (another material, a lighter design).
2. **Parts.** Each part's material, finished mass and number; its process; the share **machined** away (it adds to the material bought and charges machining); its **recycled** content (the typical share of today's supply is shown); and its **end of life**: landfill, burn for energy, downcycle, recycle or not counted.
3. **The totals:** CO_2 and energy for the first four phases, with their ranges, and the net CO_2 with the end-of-life credit.
4. **The largest phase**, and what usually helps for it.
5. **The phase bars**, CO_2 and energy, each with its range.

Below the parts, **Transport** takes legs by sea, rail, road or air with their distances, and **Use** is either **Powered** (kW, hours a day, days a year, on the electricity grid chosen at the top) or **Carried in a vehicle** (the fuel to carry the product's own mass, for the distance travelled a year), for a life in years.

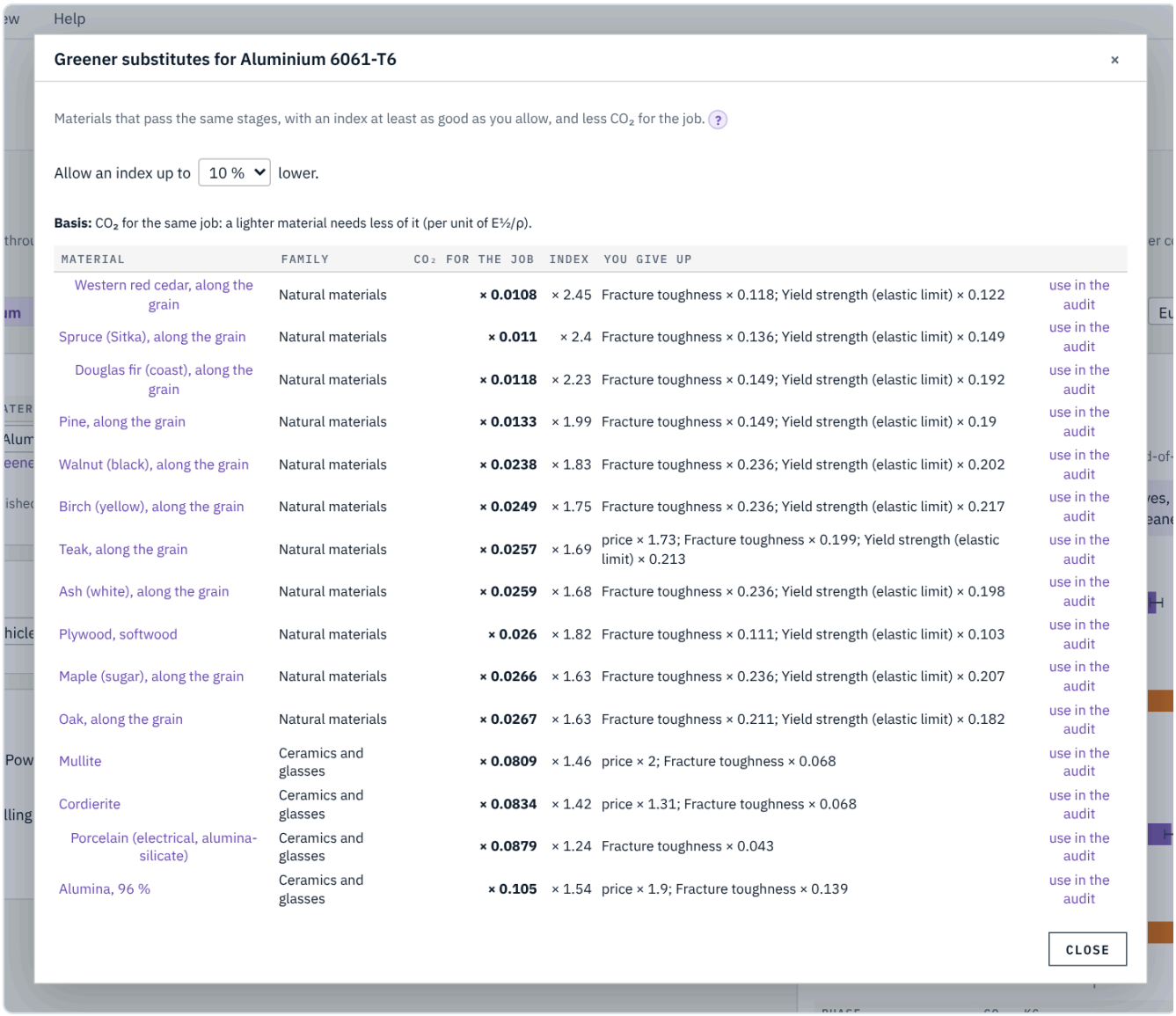


Two options side by side: where a lighter design pays back.

With two or more options, **Options side by side** compares them phase by phase, and the chart draws each option's CO_2 against years of use: what it costs to make is where its line starts, and the use phase is its slope. Where the lines cross, the options **break even**.

Every figure is a range from the values' own ranges and public factors (freight, vehicles, electricity in eleven countries and regions, process energy). A material without a value is listed in the notes, never counted as zero. This is a design-stage audit, not a full life-cycle assessment. The audit goes into the report, and **To a spreadsheet** saves every option, phase and part.

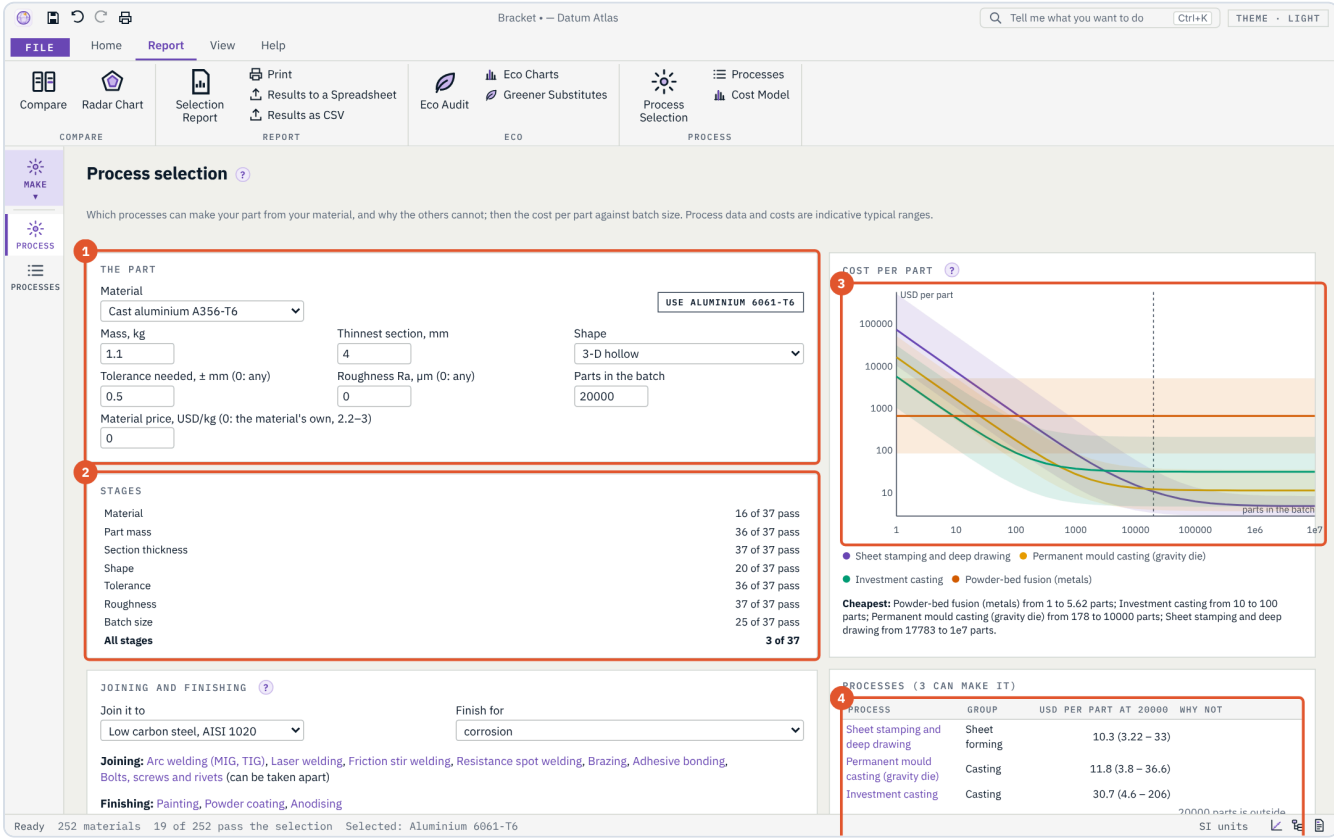
Greener substitutes



Greener substitutes for the aluminium bracket.

Greener substitutes (from a datasheet, the ribbon or **greener?** beside an audit part) lists materials that pass the same stages, with an index nearly as good (you choose how much lower is allowed), and less CO_2 **for the same job**. With a minimum-mass index such as E½/ρ, a better index means less material, so CO_2 for the job is CO_2 per kilogram divided by the index. **You give up** lists price and properties at least 20 % worse. **Use in the audit** swaps the part's material.

Process selection

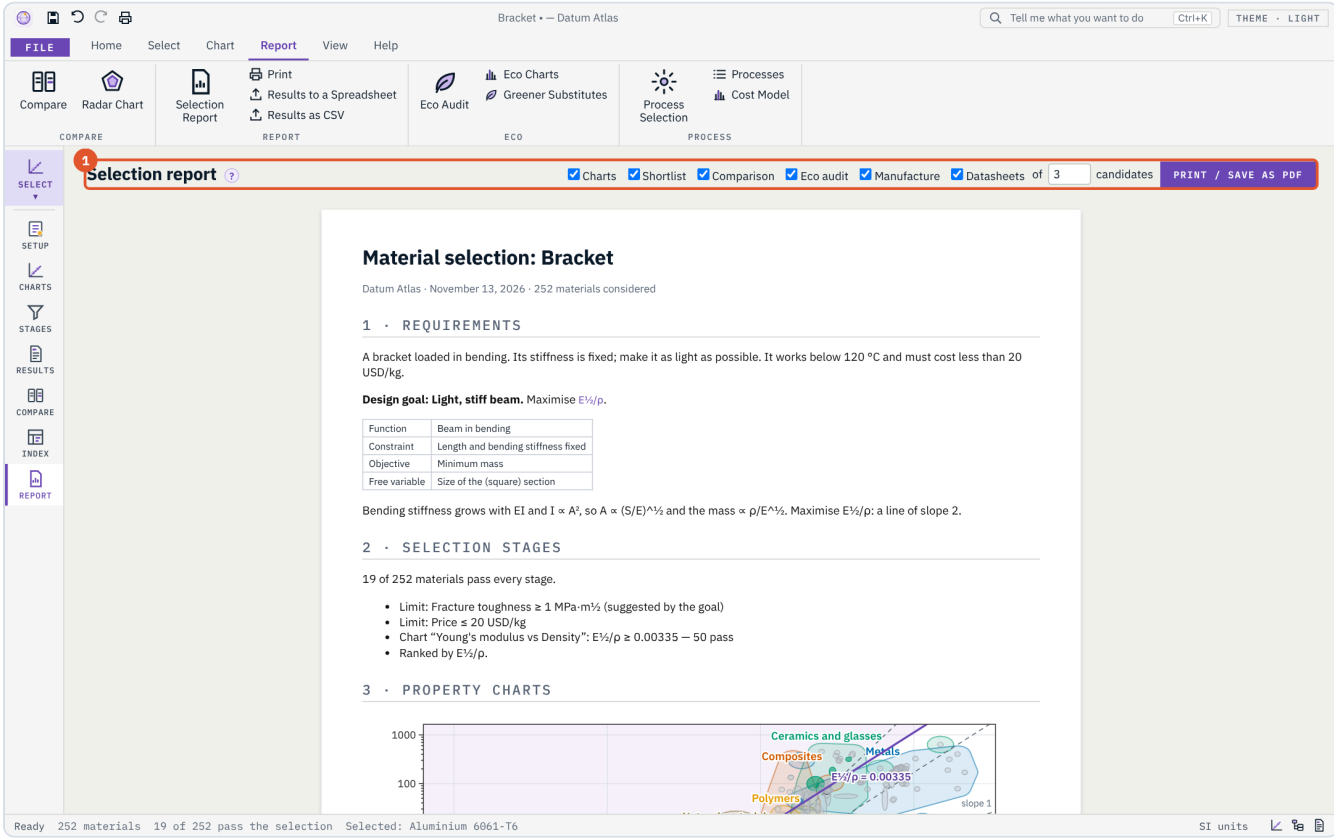


Process selection for a cast aluminium part.

1. **The part:** its material (or **Use …** for the one picked on a chart), mass, thinnest section, shape (flat, sheet, prismatic, 3-D solid, 3-D hollow), the tolerance and surface roughness it needs, the number of parts, and the material price (0 uses the material's own).
2. **Stages:** how many of the processes pass each requirement, and all of them.
3. **Cost per part** against batch size, on log axes: material (with what is wasted) + tooling shared over the batch + the machine's hourly cost divided by its rate. Tooling makes small batches dear; where the curves cross, the cheapest process changes, and the line below the chart says which is cheapest for which batch sizes. The dashed line is your batch size.
4. **Processes:** those that can make the part first, cheapest first; every other one says what stops it (*does not work this alloy*, *cannot make a 3-D hollow part*, *its best tolerance is ±0.8 mm*, *outside its economic batch*).

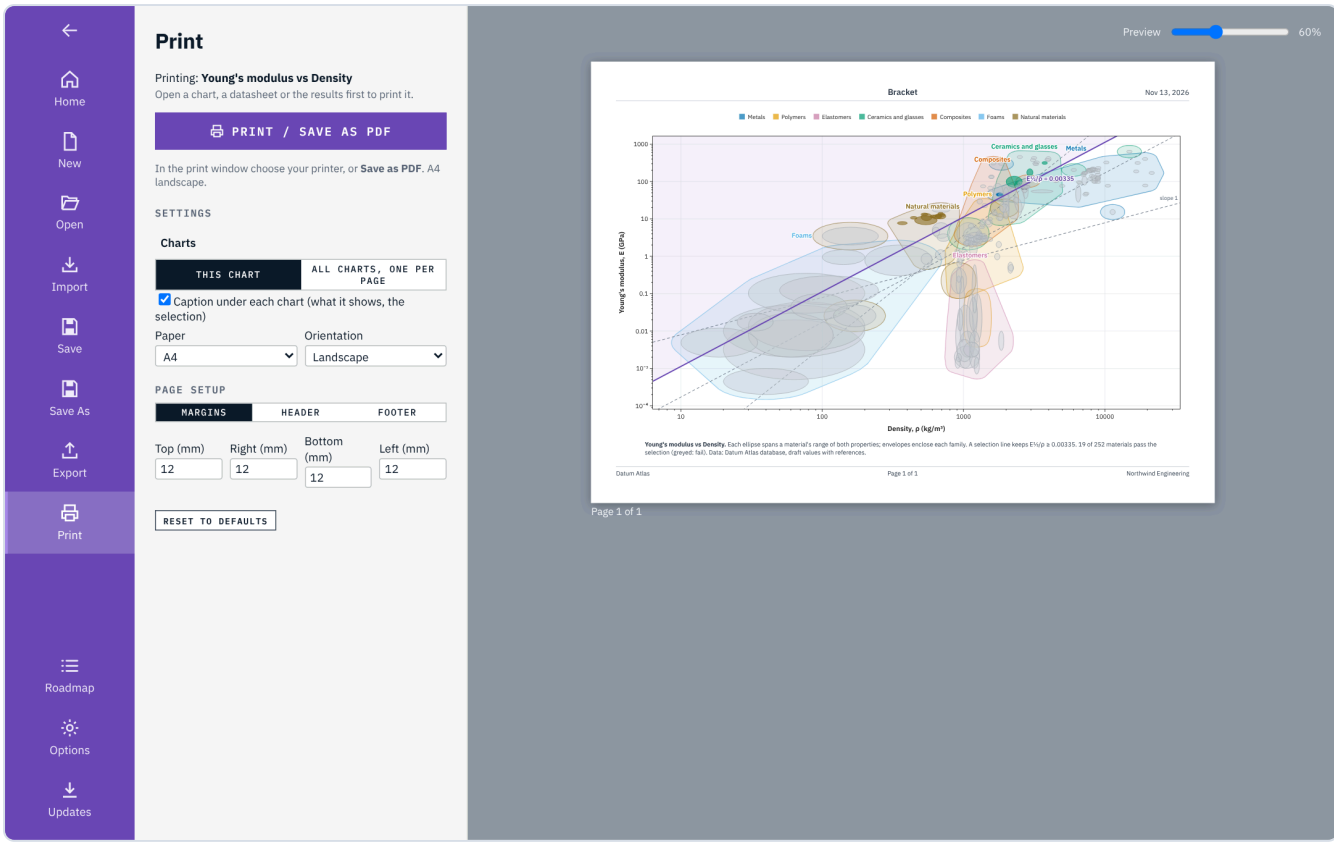
Below them: **Joining and finishing** lists ways to join the material to itself or to another, and finishes for what you need (corrosion, wear, decoration…); **Processes for the shortlist** shows which processes work the materials in Compare or the best of the results. **Report › Processes** lists every process record; click one for its card. Process data and costs are indicative typical ranges, not quotes: ask your supplier.

The report, printing and export



The selection report.

1. Choose what the report holds: the requirements and the design goal, the stages, the **charts**, the **shortlist**, the **comparison**, the **eco audit**, the **manufacture**, and the **datasheets** of the best candidates with their sources. **Print / Save as PDF** prints it.



File › Print, with the preview.

File › Print (`Ctrl+P`) prints the open chart (vector, with its legend and a caption), a datasheet, the results or the report, with page setup, margins, headers and footers. Choose **Save as PDF** in the print window for a PDF.

Other ways out:

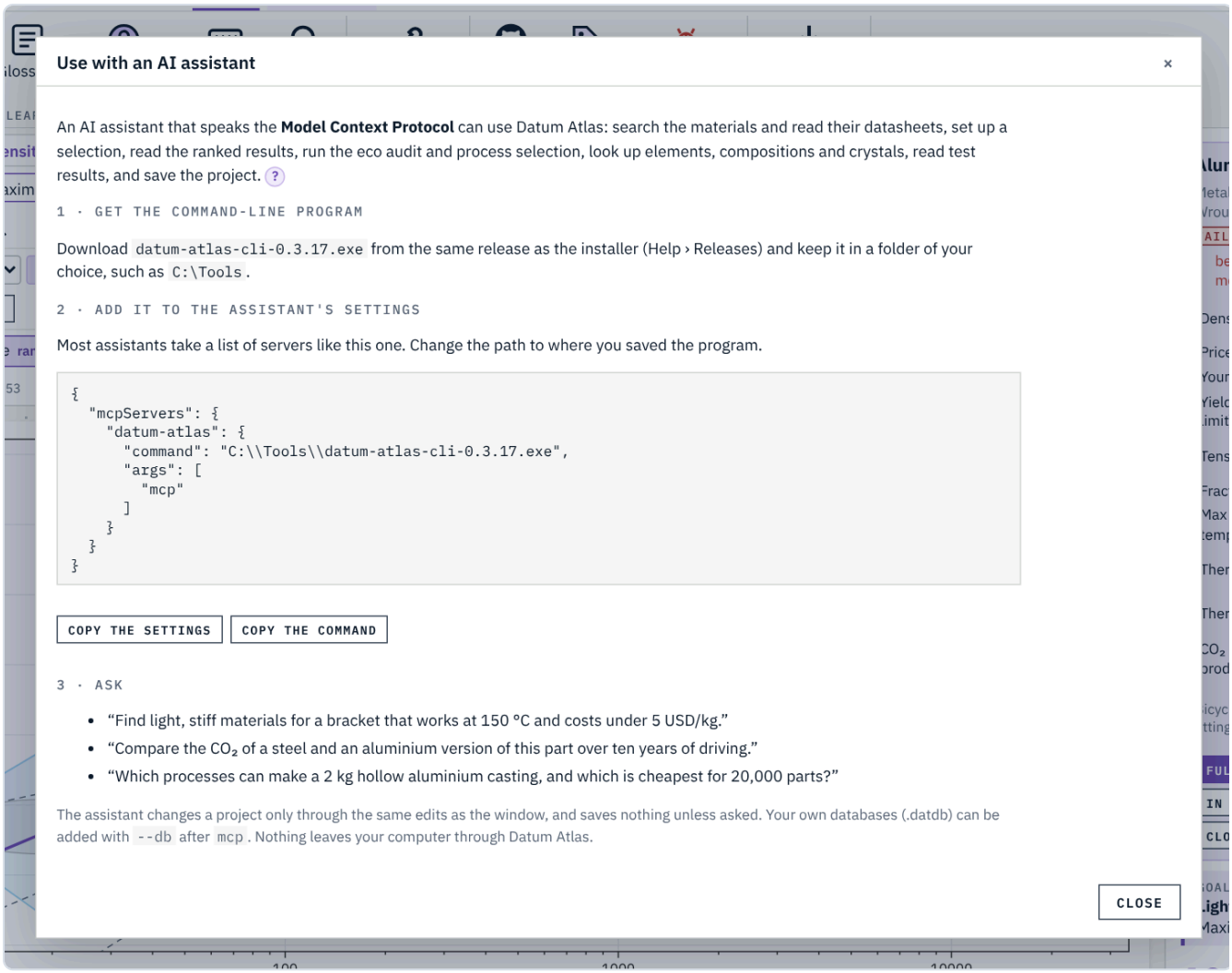
- **Chart › Export:** the chart as SVG (sharp at any size) or PNG (at the resolution in Options). **Copy Picture** puts it on the clipboard for a document or a slide.
- **Results** and the **eco audit** as `.xlsx` or `.csv`.
- **Your materials** as `.datdb`, `.xlsx` or `.csv`.

The command line and AI assistants

Every release has a small program beside the installer, `datum-atlas-cli-x.y.z.exe`. It does what the window does, from a command prompt:

Command	What it does
`search 6061`	Materials by name, designation or use
`material al-6061-t6`	A datasheet: values, units, sources, ✓
`new beam bracket.datlas`	Start a project from a design problem
`results bracket.datlas --csv`	Every material ranked, and why the others fail
`ops`	The named edits, each with an example
`apply bracket.datlas --op '…' --save new.datlas`	Change a project with a named edit
`eco`, `process`, `substitutes`	The eco audit, process selection, greener substitutes
`export bracket.datlas results.xlsx`	Save as .datlas, .xlsx or .csv

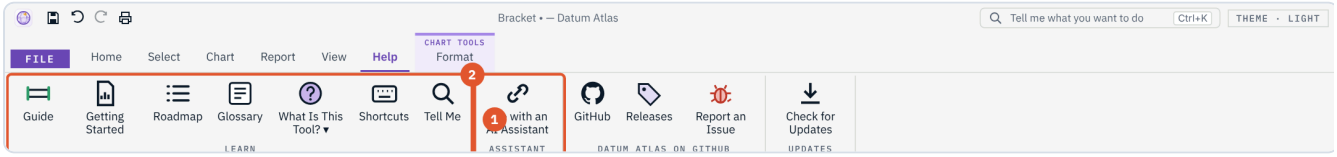
`--db mine.datdb` adds a database of your own. Nothing is saved unless you say `--save` or `export`.



Help › Use with an AI Assistant.

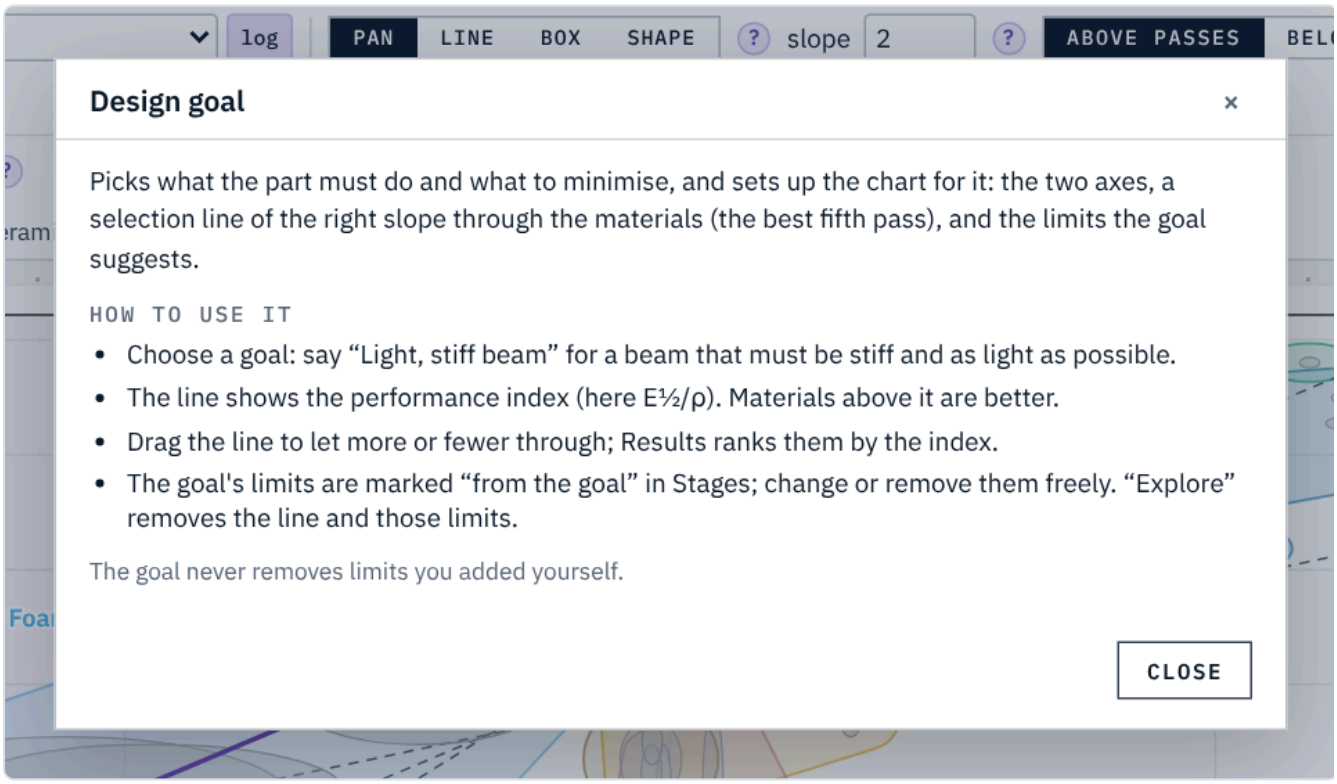
`datum-atlas-cli mcp` connects an AI assistant that speaks the Model Context Protocol. The assistant can search the materials and read their datasheets, set up a selection with the same edits as the window, read the ranked results, run the eco audit and process selection, look up elements, compositions and crystals, read test results, and save the project when you ask. **Help › Use with an AI Assistant** has the settings to copy into the assistant.

Help, options and shortcuts



The Help tab.

1. **Learn: Getting Started** (the method in one page), **Roadmap** (what each coming release brings), **Glossary** (bubble, envelope, index, × best, life phases, economic batch… with a search box), **What Is This Tool?** (every ? card), **Shortcuts** and **Tell Me**.
2. **Assistant:** how to connect an AI assistant.



A ? card.

File › Options: the theme (light, dark, or auto with Windows), SI or imperial units, autosave every few minutes, labels and guide lines on new charts, the picture resolution, paper size and orientation, and your company name and logo for printed pages.

Keys	What they do
`Ctrl+K` or `Alt+Q`	Tell Me: find a command, a goal or a material
`Ctrl+N` `Ctrl+O` `Ctrl+S`	New (File › Home), open, save
`Ctrl+Shift+S`	Save As
`Ctrl+P`	Print

Keys	What they do
`Ctrl+Z` `Ctrl+Y`	Undo, redo
Wheel, drag	Zoom and pan the chart
Arrow keys	Move the active selection line (`Shift`: ten times further)
`Ctrl+C`	Copy the selected table cells
`F1`	Keyboard shortcuts

If Datum Atlas closes before you save, it offers to recover your work the next time it starts.

> **Tip.** Report a problem or ask for a feature with **Help › Report an Issue**. Corrections to a value, with a reference, are welcome.